

TRAINING REGULATIONS

PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II



SOCIAL AND OTHER COMMUNITY DEVELOPMENT SERVICES SECTOR

TECHNICAL EDUCATION AND SKILLS DEVELOPMENT AUTHORITY

East Service Road, South Luzon Expressway (SLEX), Taguig City, Metro Manila

*Technical Education and Skills Development Act of 1994
(Republic Act No. 7796)*

Section 22, “Establishment and Administration of the National Trade Skills Standards” of the RA 7796 known as the TESDA Act mandates TESDA to establish national occupational skill standards. The Authority shall develop and implement a certification and accreditation program in which private industry group and trade associations are accredited to conduct approved trade tests, and the local government units to promote such trade testing activities in their respective areas in accordance with the guidelines to be set by the Authority.

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The Training Regulations (TR) serve as basis for the:

- 1 Development of curriculum and assessment instruments;
- 2 Registration and delivery of training programs; and
- 3 Competency assessment and certification

Each TR has four sections:

- Section 1 **Definition of Qualification** describes the qualification and defines the competencies that comprise the qualification.
- Section 2 **Competency Standards** gives the specifications of competencies required for effective work performance.
- Section 3 **Training Arrangements** contains information and requirements in designing training program for certain qualification. It includes curriculum design; training delivery; trainee entry requirements; tools, equipment and materials; training facilities; trainer's qualification; and institutional assessment.
- Section 4 **Assessment and Certification Arrangements** describes the policies governing assessment and certification procedures.

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TRAINING REGULATIONS FOR PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II

SECTION 1 DEFINITION OF QUALIFICATION

The **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II** Qualification consists of competencies that a person must achieve to perform halftone reproduction and its requirements, perform basic color separation process, perform digital output processing operations, perform imposition procedures and perform regular maintenance of condition-based prepress machines/equipment.

The Units of Competency comprising this Qualification include the following:

Unit Code	BASIC COMPETENCIES
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400311210	Participate in workplace communication
400311211	Work in a team environment
400311212	Solve/address general workplace problems
400311213	Develop career and life decisions
400311214	Contribute to workplace innovation
400311215	Present relevant information
400311216	Practice occupational safety and health policies and procedures
400311217	Exercise efficient and effective sustainable practices in the workplace
400311218	Practice entrepreneurial skills in the workplace

Unit Code	COMMON COMPETENCIES
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SOC532201	Update industry knowledge
SOC514202	Manage own performance
SOC514204	Maintain safe, clean and efficient work environment
ELC311203	Perform basic computer operations

Unit Code	CORE COMPETENCIES
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SOC732306	Perform halftone reproduction and its requirements
SOC732307	Perform basic color separation process
SOC732308	Perform digital output processing operations
SOC732309	Perform imposition procedures
SOC732310	Perform regular maintenance of condition-based prepress machines/equipment

A person who has achieved this Qualification is competent to be:

- ☐ Senior Prepress Technician

SECTION 2 COMPETENCY STANDARDS

This section gives the details of the contents of the units of competency required in **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II**.

BASIC COMPETENCIES

UNIT OF COMPETENCY : PARTICIPATE IN WORKPLACE COMMUNICATION

UNIT CODE : 400311210

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to gather, interpret and convey information in response to workplace requirements.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Obtain and convey workplace information	1.1 Specific and relevant information is accessed from <i>appropriate sources</i>. 1.2 Effective questioning, active listening and speaking skills are used to gather and convey information. 1.3 Appropriate <i>medium</i> is used to transfer information and ideas. 1.4 Appropriate non- verbal communication is used. 1.5 Appropriate lines of communication with supervisors and colleagues are identified and followed. 1.6 Defined workplace procedures for the location and <i>storage</i> of information are used. 1.7 Personal interaction is carried out clearly and concisely.	1.1 Effective verbal and nonverbal communication 1.2 Different modes of communication 1.3 Medium of communication in the workplace 1.4 Organizational policies 1.5 Communication procedures and systems 1.6 Lines of Communication 1.7 Technology relevant to the enterprise and the individual's work responsibilities 1.8 Workplace etiquette	1.1 Following simple spoken language 1.2 Performing routine workplace duties following simple written notices 1.3 Participating in workplace meetings and discussions 1.4 Preparing work-related documents 1.5 Estimating, calculating and recording routine workplace measures 1.6 Relating/ Interacting with people of various levels in the workplace 1.7 Gathering and providing basic information in response to workplace requirements

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
			1.8 Applying basic business writing skills 1.9 Applying interpersonal skills in the workplace 1.10 Performing active-listening skills
2. Perform duties following workplace instructions	2.1 Written notices and instructions are read and interpreted in accordance with organizational guidelines. 2.2 Routine written instruction is followed based on established procedures. 2.3 Feedback is given to workplace supervisor-based instructions/ information received. 2.4 Workplace interactions are conducted in a courteous manner. 2.5 Where necessary, clarifications about routine workplace procedures and matters concerning conditions of employment are sought and asked from appropriate sources . 2.6 Meeting outcomes are interpreted and implemented.	2.1 Effective verbal and non-verbal communication 2.2 Different modes of communication 2.3 Medium of communication in the workplace 2.4 Organizational/ workplace policies 2.5 Communication procedures and systems 2.6 Lines of communication 2.7 Technology relevant to the enterprise and the individual's work responsibilities 2.8 Effective questioning techniques (clarifying and probing) 2.9 Workplace etiquette	2.1 Following simple spoken instructions 2.2 Performing routine workplace duties following simple written notices 2.3 Participating in workplace meetings and discussions 2.4 Completing work-related documents 2.5 Estimating, calculating and recording routine workplace measures 2.6 Relating/ Responding to people of various levels in the workplace 2.7 Gathering and providing information in response to workplace requirements 2.8 Applying basic questioning/ querying 2.9 Applying skills in reading for information 2.10 Applying skills in locating

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Complete relevant work related documents	3.1 Range of forms relating to conditions of employment are completed accurately and legibly. 3.2 Workplace data is recorded on standard workplace forms and documents. 3.3 Errors in recording information on forms/ documents are identified and acted upon. 3.4 Reporting requirements to supervisor are completed according to organizational guidelines.	3.1 Effective verbal and non-verbal communication 3.2 Different modes of communication 3.3 Workplace forms and documents 3.4 Organizational/ Workplace policies 3.5 Communication procedures and systems 3.6 Technology relevant to the enterprise and the individual's work responsibilities	3.1 Completing work-related documents 3.2 Applying operations of addition, subtraction, division and multiplication 3.3 Gathering and providing information in response to workplace requirements 3.4 Applying effective record keeping skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Appropriate sources	May include: 1.1. Team members 1.2. Supervisor/Department Head 1.3. Suppliers 1.4. Trade personnel 1.5. Local government 1.6. Industry bodies
2. Medium	May include: 2.1. Memorandum 2.2. Circular 2.3. Notice 2.4. Information dissemination 2.5. Follow-up or verbal instructions 2.6. Face-to-face communication 2.7. Electronic media (disk files, cyberspace)
3. Storage	May include: 3.1. Manual filing system 3.2. Computer-based filing system
4. Workplace interactions	May include: 4.1. Face-to-face 4.2. Telephone 4.3. Electronic and two-way radio 4.4. Written including electronic means, memos, instruction and forms 4.5. Non-verbal including gestures, signals, signs and diagrams
5. Forms	May include: 5.1. HR/Personnel forms, telephone message forms, safety reports

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared written communication following standard format of the organization 1.2 Accessed information using workplace communication equipment/systems 1.3 Made use of relevant terms as an aid to transfer information effectively 1.4 Conveyed information effectively adopting formal or informal communication
2. Resource Implications	The following resources should be provided: 2.1 Fax machine 2.2 Telephone 2.3 Notebook 2.4 Writing materials 2.5 Computer with Internet connection
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with oral questioning 3.2 Interview 3.3 Written test 3.4 Third-party report
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or through an accredited institution

UNIT OF COMPETENCY : WORK IN A TEAM ENVIRONMENT

UNIT CODE : 400311211

UNIT DESCRIPTOR : This unit covers the skills, knowledge and attitudes to identify one's roles and responsibilities as a member of a team.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Describe team role and scope	1.1 The <i>role and objective of the team</i> is identified from available <i>sources of information</i> . 1.2 Team parameters, reporting relationships and responsibilities are identified from team discussions and appropriate external sources.	1.1 Group structure 1.2 Group development 1.3 Sources of information	1.1 Communicating with others, appropriately consistent with the culture of the workplace 1.2 Developing ways in improving work structure and performing respective roles in the group or organization
2. Identify one's role and responsibility within a team	2.1 Individual roles and responsibilities within the team environment are identified. 2.2 Roles and objectives of the team is identified from available <i>sources of information</i> . 2.3 Team parameters, reporting relationships and responsibilities are identified based on team discussions and appropriate external sources.	2.1 Team roles and objectives 2.2 Team structure and parameters 2.3 Team development 2.4 Sources of information	2.1 Communicating with others, appropriately consistent with the culture of the workplace 2.2 Developing ways in improving work structure and performing respective roles in the group or organization
3. Work as a team member	3.1 Effective and appropriate forms of communications are used and interactions undertaken with team members based on company practices. 3.2 Effective and appropriate contributions made to complement team activities and objectives,	3.1 Communication Process 3.2 Workplace communication protocol 3.3 Team planning and decision making 3.4 Team thinking	3.1 Communicating appropriately, consistent with the culture of the workplace 3.2 Interacting effectively with others 3.3 Deciding as an individual and as

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	based on <i>workplace context.</i> 3.3 Protocols in reporting are observed based on standard company practices. 3.4 Contribute to the development of team work plans based on an understanding of team's role and objectives.	3.5 Team roles 3.6 Process of team development 3.7 Workplace context	a group using group think strategies and techniques 3.4 Contributing to Resolution of issues and concerns

RANGE OF VARIABLES

VARIABLE	RANGE
1. Role and objective of team	May include but not limited to: 1.1 Work activities in a team environment with enterprise or specific sector 1.2 Limited discretion, initiative and judgement maybe demonstrated on the job, either individually or in a team environment
2. Sources of information	May include but not limited to: 2.1 Standard operating and/or other workplace procedures 2.2 Job procedures 2.3 Machine/equipment manufacturer's specifications and instructions 2.4 Organizational or external personnel 2.5 Client/supplier instructions 2.6 Quality standards 2.7 OHS and environmental standards
3. Workplace context	May include but not limited to: 3.1 Work procedures and practices 3.2 Conditions of work environments 3.3 Legislation and industrial agreements 3.4 Standard work practice including the storage, safe handling and disposal of chemicals 3.5 Safety, environmental, housekeeping and quality guidelines

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Worked in a team to complete workplace activity 1.2 Worked effectively with others 1.3 Conveyed information in written or oral form 1.4 Selected and used appropriate workplace language 1.5 Followed designated work plan for the job
2. Resource Implications	The following resources should be provided: 2.1 Access to relevant workplace or appropriately simulated environment where assessment can take place 2.2 Materials relevant to the proposed activity or tasks
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Role play involving the participation of individual member to the attainment of organizational goal 3.2 Case studies and scenarios as a basis for discussion of issues and strategies in teamwork 3.3 Socio-drama and socio-metric methods 3.4 Sensitivity techniques 3.5 Written test
4. Context for Assessment	4.1 Competency may be assessed in workplace or in a simulated workplace setting 4.2 Assessment shall be observed while task are being undertaken whether individually or in group

UNIT OF COMPETENCY : SOLVE/ADDRESS GENERAL WORKPLACE PROBLEMS

UNIT CODE : 400311212

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to apply problem-solving techniques to determine the origin of problems and plan for their resolution. It also includes addressing procedural problems through documentation, and referral.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify routine problems	1.1 Routine <i>problems or procedural problem</i> areas are identified. 1.2 Problems to be investigated are defined and determined. 1.3 Current conditions of the problem are identified and documented.	1.1 Current industry hardware and software products and services 1.2 Industry maintenance, service and helpdesk practices, processes and procedures 1.3 Industry standard diagnostic tools 1.4 Malfunctions and resolutions	1.1 Identifying current industry hardware and software products and services 1.2 Identifying current industry maintenance, services and helpdesk practices, processes and procedures. 1.3 Identifying current industry standard diagnostic tools 1.4 Describing common malfunctions and resolutions. 1.5 Determining the root cause of a routine malfunction

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Look for solutions to routine problems	2.1 Potential solutions to problem are identified. 2.2 Recommendations about possible solutions are developed, documented, ranked and presented to appropriate person for decision.	2.1 Current industry hardware and software products and services 2.2 Industry service and helpdesk practices, processes and procedures 2.3 Operating systems 2.4 Industry standard diagnostic tools 2.5 Malfunctions and resolutions. 2.6 Root cause analysis	2.1 Identifying current industry hardware and software products and services 2.2 Identifying services and helpdesk practices, processes and procedures. 2.3 Identifying operating system 2.4 Identifying current industry standard diagnostic tools 2.5 Describing common malfunctions and resolutions. 2.6 Determining the root cause of a routine malfunction
3. Recommend solutions to problems	3.1 Implementation of solutions are planned. 3.2 Evaluation of implemented solutions are planned. 3.3 Recommended solutions are documented and submit to appropriate person for confirmation.	3.1 Standard procedures 3.2 Documentation produce	3.1 Producing documentation that recommends solutions to problems 3.2 Following established procedures

RANGE OF VARIABLES

VARIABLE	RANGE
1. Problems/Procedural Problem	May include but not limited to: 1.1 Routine/non – routine processes and quality problems 1.2 Equipment selection, availability and failure 1.3 Teamwork and work allocation problem 1.4 Safety and emergency situations and incidents 1.5 Work-related problems outside of own work area
2. Appropriate person	May include but not limited to: 2.1 Supervisor or manager 2.2 Peers/work colleagues 2.3 Other members of the organization
3. Document	May include but not limited to: 3.1 Electronic mail 3.2 Briefing notes 3.3 Written report 3.4 Evaluation report
4. Plan	May include but not limited to: 4.1 Priority requirements 4.2 Co-ordination and feedback requirements 4.3 Safety requirements 4.4 Risk assessment 4.5 Environmental requirements

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Determined the root cause of a routine problem 1.2 Identified solutions to procedural problems. 1.3 Produced documentation that recommends solutions to problems. 1.4 Followed established procedures. 1.5 Referred unresolved problems to support persons.
2. Resource Implications	2.1. Assessment will require access to a workplace over an extended period, or a suitable method of gathering evidence of operating ability over a range of situations.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Case Formulation 3.2 Life Narrative Inquiry 3.3 Standardized test The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components.
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions.

UNIT OF COMPETENCY : DEVELOP CAREER AND LIFE DECISIONS

UNIT CODE : 400311213

UNIT DESCRIPTOR : This unit covers the knowledge, skills, and attitudes in managing one's emotions, developing reflective practice, and boosting self-confidence and developing self-regulation.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Manage one's emotion	1.1 <i>Self-management strategies</i> are identified. 1.2 Skills to work independently and to show initiative, to be conscientious, and persevering in the face of setbacks and frustrations are developed. 1.3 Techniques for effectively handling negative emotions and <i>unpleasant situation</i> in the workplace are examined.	1.1 Self-management strategies that assist in regulating behavior and achieving personal and learning goals (e.g. Nine self-management strategies according to Robert Kelley) 1.2 Enablers and barriers in achieving personal and career goals 1.3 Techniques in handling negative emotions and unpleasant situation in the workplace such as frustration, anger, worry, anxiety, etc.	1.1 Managing properly, one's emotions and recognizing situations that cannot be changed and accept them and remain professional 1.2 Developing self-discipline, working independently and showing initiative to achieve personal and career goals 1.3 Showing confidence, and resilience in the face of setback and frustrations and other negative emotions and unpleasant situations in the workplace
2. Develop reflective practice	2.1 Personal strengths and achievements, based on self-assessment strategies and teacher feedback are contemplated. 2.2 Progress when seeking and responding to feedback from teachers to assist them in consolidating strengths, addressing weaknesses	2.1 Basic SWOT analysis 2.2 Strategies to improve one's attitude in the workplace 2.3 Gibbs' Reflective Cycle/Model (Description, Feelings, Evaluation, Analysis,	2.1 Using the basic SWOT analysis as self-assessment strategy 2.2 Developing reflective practice through realization of limitations, likes/dislikes; through showing of self-confidence

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	and fulfilling their potential are monitored. 2.3 Outcomes of personal and academic challenges by reflecting on previous problem solving and decision-making strategies and feedback from peers and teachers are predicted.	Conclusion, and Action plan)	2.3 Demonstrating self-acceptance and being able to accept challenges
3. Boost self-confidence and develop self-regulation	3.1 Efforts for continuous self-improvement are demonstrated. 3.2 Counter-productive tendencies at work are eliminated. 3.3 Positive outlook in life are maintained.	3.1 Four components of self-regulation based on Self-Regulation Theory (SRT) 3.2 Personality development concepts 3.3 Self-help concepts (e. g., 7 Habits by Stephen Covey, transactional analysis, psycho-spiritual concepts)	3.1 Performing effective communication skills – reading, writing, conversing skills 3.2 Showing affective skills – flexibility, adaptability, etc. 3.3 Self-assessment for determining one's strengths and weaknesses

RANGE OF VARIABLES

VARIABLE	RANGE
1. Self-management strategies	May include but not limited to: 1.1 Seeking assistance in the form of job coaching or mentoring 1.2 Continuing dialogue to tackle workplace grievances 1.3 Collective negotiation/bargaining for better working conditions 1.4 Share your goals to improve with a trusted co-worker or supervisor 1.5 Make a negativity log of every instance when you catch yourself complaining to others 1.6 Make lists and schedules for necessary activities
2. Unpleasant situation	May include but not limited to: 2.1 Job burn-out 2.2 Drug dependence 2.3 Sulking

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Expressed emotions appropriately 1.2 Worked independently and show initiative 1.3 Consistently demonstrated self-confidence and self-discipline
2. Resource Implications	The following resources should be provided: 2.1. Access to workplace and resource s 2.2. Case studies
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Demonstration or simulation with oral questioning 3.2. Case problems involving work improvement and sustainability issues 3.3. Third-party report
4. Context for Assessment	4.1. Competency assessment may occur in workplace or any appropriately simulated environment

UNIT OF COMPETENCY : CONTRIBUTE TO WORKPLACE INNOVATION

UNIT CODE : 400311214

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to make a pro-active and positive contribution to workplace innovation.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify opportunities to do things better	1.1 <i>Opportunities for improvement</i> are identified proactively in own area of work. 1.2 <i>Information</i> are gathered and reviewed which may be relevant to ideas and which might assist in gaining support for idea.	1.1 Roles of individuals in suggesting and making improvements 1.2 Positive impacts and challenges in innovation 1.3 Types of changes and responsibility 1.4 Seven habits of highly effective people	1.1 Identifying opportunities to improve and to do things better involvement 1.2 Identifying the positive impacts and the challenges of change and innovation 1.3 Identifying examples of the types of changes that are within and outside own scope of responsibility
2. Discuss and develop ideas with others	2.1 <i>People who could provide input</i> to ideas for improvements are identified. 2.2 Ways of approaching people to begin sharing ideas are selected. 2.3 Meeting is set with relevant people. 2.4 Ideas for follow up are review and selected based on feedback. 2.5 <i>Critical inquiry method</i> is used to discuss and develop ideas with others.	2.1 Roles of individuals in suggesting and making improvements 2.2 Positive impacts and challenges in innovation 2.3 Types of changes and responsibility 2.4 Seven habits of highly effective people	2.1 Identifying opportunities to improve and to do things better. Involvement 2.2 Identifying the positive impacts and the challenges of change and innovation 2.3 Providing examples of the types of changes that are within and outside own scope of responsibility 2.4 Communicating ideas for change through small group discussions and meetings

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Integrate ideas for change in the workplace	3.1 Critical inquiry method is used to integrate different ideas for change of key people. 3.2 Summarizing, analyzing and generalizing skills are used to extract salient points in the pool of ideas. 3.3 Reporting skills are likewise used to communicate results. 3.4 Current Issues and concerns on the systems, processes and procedures, as well as the need for simple innovative practices are identified.	3.1 Roles of individuals in suggesting and making improvements 3.2 Positive impacts and challenges in innovation 3.3 Types of changes and responsibility 3.4 Seven habits of highly effective people 3.5 Basic research skills	3.1 Identifying opportunities to improve and to do things better involvement 3.2 Identifying the positive impacts and the challenges of change and innovation 3.3 Providing examples of the types of changes that are within and outside own scope of responsibility 3.4 Communicating ideas for change through small group discussions and meetings 3.5 Demonstrating skills in analysis and interpretation of data

RANGE OF VARIABLES

VARIABLE	RANGE
1. Opportunities for improvement	May include: 1.1 Systems 1.2 Processes 1.3 Procedures 1.4 Protocols 1.5 Codes 1.6 Practices
2. Information	May include: 2.1 Workplace communication problems 2.2 Performance evaluation results 2.3 Team dynamics issues and concerns 2.4 Challenges on return of investment 2.5 New tools, processes and procedures 2.6 New people in the organization
3. People who could provide input	May include: 3.1 Leaders 3.2 Managers 3.3 Specialists 3.4 Associates 3.5 Researchers 3.6 Supervisors 3.7 Staff 3.8 Consultants (external) 3.9 People outside the organization in the same field or similar expertise/industry 3.10 Clients
4. Critical inquiry method	May include: 4.1 Preparation 4.2 Discussion 4.3 Clarification of goals 4.4 Negotiate towards a Win-Win outcome 4.5 Agreement 4.6 Implementation of a course of action 4.7 Effective verbal communication. See our pages Verbal Communication and Effective Speaking 4.8 Listening 4.9 Reducing misunderstandings is a key part of effective negotiation 4.10 Rapport Building 4.11 Problem Solving 4.12 Decision Making 4.13 Assertiveness 4.14 Dealing with Difficult Situations

VARIABLE	RANGE
5. Reporting skills	May include: 5.1 Data management 5.2 Coding 5.3 Data analysis and interpretation 5.4 Coherent writing 5.5 Speaking

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified opportunities to do things better 1.2 Discussed and developed ideas with others on how to contribute to workplace innovation 1.3 Integrated ideas for change in the workplace 1.4 Analyzed and reported rooms for innovation and learning in the workplace
2. Resource Implications	The following resources should be provided: 2.1 Pens, papers and writing implements 2.2 White board 2.3 Manila papers
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Psychological and behavioral Interviews 3.2 Performance Evaluation 3.3 Life Narrative Inquiry 3.4 Review of portfolios of evidence and third-party workplace reports of on-the-job performance. 3.5 Sensitivity analysis 3.6 Organizational analysis 3.7 Standardized assessment of character strengths and virtues applied
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : PRESENT RELEVANT INFORMATION

UNIT CODE : 400311215

UNIT DESCRIPTOR : This unit of covers the knowledge, skills and attitudes required to present data/information appropriately.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Gather data/information	1.1 Evidence, facts and information are collected. 1.2 Evaluation, terms of reference and conditions are reviewed to determine whether data/information falls within project scope.	1.1 Organisational protocols 1.2 Confidentiality 1.3 Accuracy 1.4 Business mathematics and statistics 1.5 Data analysis techniques/procedures 1.6 Reporting requirements to a range of audiences 1.7 Legislation, policy and procedures relating to the conduct of evaluations 1.8 Organisational values, ethics and codes of conduct	1.1 Describing organisational protocols relating to client liaison 1.2 Protecting confidentiality 1.3 Describing accuracy 1.4 Computing business mathematics and statistics 1.5 Describing data analysis techniques/procedures 1.6 Reporting requirements to a range of audiences 1.7 Stating legislation, policy and procedures relating to the conduct of evaluations 1.8 Stating organisational values, ethics and codes of conduct

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Assess gathered data/information	2.1 Validity of data/information is assessed. 2.2 Analysis techniques are applied to assess data/information. 2.3 Trends and anomalies are identified. 2.4 Data analysis techniques and procedures are documented. 2.5 Recommendations are made on areas of possible improvement.	2.1 Business mathematics and statistics 2.2 Data analysis techniques/procedures 2.3 Reporting requirements to a range of audiences 2.4 Legislation, policy and procedures relating to the conduct of evaluations 2.5 Organisational values, ethics and codes of conduct	2.1 Computing business mathematics and statistics 2.2 Describing data analysis techniques/procedures 2.3 Reporting requirements to a range of audiences 2.4 Stating legislation, policy and procedures relating to the conduct of evaluations 2.5 Stating organisational values, ethics and codes of conduct
3. Record and present information	3.1 Studied data/information are recorded. 3.2 Recommendations are analysed for action to ensure they are compatible with the project's scope and terms of reference. 3.3 Interim and final reports are analysed and outcomes are compared to the criteria established at the outset. 3.4 Findings are presented to stakeholders.	3.1 Data analysis techniques/procedures 3.2 Reporting requirements to a range of audiences 3.3 Legislation, policy and procedures relating to the conduct of evaluations 3.4 Organisational values, ethics and codes of conduct	3.1 Describing data analysis techniques/procedures 3.2 Reporting requirements to a range of audiences 3.3 Stating legislation, policy and procedures relating to the conduct of evaluations 3.4 Stating organisational values, ethics and codes of conduct practices

RANGE OF VARIABLES

VARIABLE	RANGE
1. Data analysis techniques	May include but not limited to: 1.1. Domain analysis 1.2. Content analysis 1.3. Comparison technique

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Determine data / information 1.2 Studied and applied gathered data/information 1.3 Recorded and studied data/information These aspects may be best assessed using a range of scenarios what ifs as a stimulus with a walk through forming part of the response. These assessment activities should include a range of problems, including new, unusual and improbable situations that may have happened.
2. Resource Implications	Specific resources for assessment 2.1 Evidence of competent performance should be obtained by observing an individual in an information management role within the workplace or operational or simulated environment.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Written Test 3.2 Interview 3.3 Portfolio The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components.
4. Context for Assessment	4.1 In all workplace, it may be appropriate to assess this unit concurrently with relevant teamwork or operation units.

UNIT OF COMPETENCY : PRACTICE OCCUPATIONAL SAFETY AND HEALTH POLICIES AND PROCEDURES

UNIT CODE : 400311216

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to identify OSH compliance requirements, prepare OSH requirements for compliance, perform tasks in accordance with relevant OSH policies and procedures.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify OSH compliance requirements	1.1 Relevant OSH requirements, regulations, policies and procedures are identified in accordance with workplace policies and procedures. 1.2 OSH activity non-conformities are conveyed to appropriate personnel . 1.3 OSH preventive and control requirements are identified in accordance with OSH work policies and procedures.	1.1 OSH preventive and control requirements 1.2 Hierarchy of Controls 1.3 Hazard Prevention and Control 1.4 General OSH principles 1.5 Work standards and procedures 1.6 Safe handling procedures of tools, equipment and materials 1.7 Standard emergency plan and procedures in the workplace	1.1 Applying communication skills 1.2 Applying interpersonal skills 1.3 Applying critical thinking skills 1.4 Applying observation skills
2. Prepare OSH requirements for compliance	2.1 OSH work activity material, tools and equipment requirements are identified in accordance with workplace policies and procedures. 2.2 Required OSH materials, tools and equipment are acquired in accordance with workplace policies and procedures. 2.3 Required OSH materials, tools and equipment are arranged/ placed in accordance with OSH work standards.	2.1 Resources necessary to execute hierarchy of controls 2.2 General OSH principles 2.3 Work standards and procedures 2.4 Safe handling procedures of tools, equipment and materials 2.5 Different OSH control measures	2.1 Applying Communication skills 2.2 Applying estimation skills 2.3 Applying interpersonal skills 2.4 Applying critical thinking skills 2.5 Applying observation skills 2.6 Identifying material, tool and equipment

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Perform tasks in accordance with relevant OSH policies and procedures	3.1 Relevant OSH work procedures are identified in accordance with workplace policies and procedures. 3.2 Work Activities are executed in accordance with OSH work standards. 3.3 <i>Non-compliance work activities</i> are reported to <i>appropriate personnel.</i>	3.1 OSH work standards 3.2 Industry related work activities 3.3 General OSH principles 3.4 OSH Violations 3.5 Non-compliance work activities	3.1 Applying communication skills 3.2 Applying interpersonal skills 3.3 Applying troubleshooting skills 3.4 Applying critical thinking skills 3.5 Applying observation skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. OSH Requirements, Regulations, Policies and Procedures	May include: 1.1 Clean Air Act 1.2 Building code 1.3 National Electrical and Fire Safety Codes 1.4 Waste management statutes and rules 1.5 Permit to Operate 1.6 Philippine Occupational Safety and Health Standards 1.7 Department Order No. 13 (Construction Safety and Health) 1.8 ECC regulations
2. Appropriate Personnel	May include: 2.1 Manager 2.2 Safety Officer 2.3 EHS Offices 2.4 Supervisors 2.5 Team Leaders 2.6 Administrators 2.7 Stakeholders 2.8 Government Official 2.9 Key Personnel 2.10 Specialists 2.11 Himself
3. OSH Preventive and Control Requirements	May include: 3.1 Resources needed for removing hazard effectively 3.2 Resources needed for substitution or replacement 3.3 Resources needed to establishing engineering controls 3.4 Resources needed for enforcing administrative controls 3.5 Personal Protective equipment
4. Non OSH-Compliance Work Activities	May include non-compliance or observance of the following safety measures: 4.1 Violations that may lead to serious physical harm or death 4.2 Fall Protection 4.3 Hazard Communication 4.4 Respiratory Protection 4.5 Power Industrial Trucks 4.6 Lockout/Tag-out 4.7 Working at heights (use of ladder, scaffolding) 4.8 Electrical Wiring Methods

	4.9 Machine Guarding 4.10 Electrical General Requirements 4.11 Asbestos work requirements 4.12 Excavations work requirements
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EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Conveyed OSH work non-conformities to appropriate personnel 1.2 Identified OSH preventive and control requirements 1.3 Identified OSH work activity material, tools and equipment requirements 1.4 Arranged/Placed required OSH materials, tools and equipment 1.5 Executed work activities 1.6 Reported OSH activity non-compliance work activities to appropriate personnel
2. Resource Implications	The following resources should be provided: 2.1 Facilities, materials tools and equipment necessary for the activity
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Observation/Demonstration with oral questioning 3.2 Third party report
4. Context for Assessment	4.1 Competency may be assessed in the work place or in a simulated work place setting

UNIT OF COMPETENCY : EXERCISE EFFICIENT AND EFFECTIVE SUSTAINABLE PRACTICES IN THE WORKPLACE

UNIT CODE : 400311217

UNIT DESCRIPTOR : This unit covers knowledge, skills and attitude to identify the efficiency and effectiveness of resource utilization, determine causes of inefficiency and/or ineffectiveness of resource utilization and convey inefficient and ineffective environmental practices.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify the efficiency and effectiveness of resource utilization	1.1 Required resource utilization in the workplace is measured using appropriate techniques. 1.2 Data are recorded in accordance with workplace protocol. 1.3 Recorded data are compared to determine the efficiency and effectiveness of resource utilization according to established <i>environmental work procedures.</i>	1.1 Importance of environmental literacy 1.2 Environmental work procedures 1.3 Waste minimization 1.4 Efficient energy consumptions	1.1 Recording skills 1.2 Writing skills 1.3 Applying innovation skills
2. Determine causes of inefficiency and/or ineffectiveness of resource utilization	2.1 Potential causes of inefficiency and/or ineffectiveness are listed. 2.2 Causes of inefficiency and/or ineffectiveness are identified through deductive reasoning. 2.3 Identified causes of inefficiency and/or ineffectiveness are validated thru established environmental procedures.	2.1 Causes of environmental inefficiencies and ineffectiveness	2.1 Applying deductive reasoning skills 2.2 Applying critical thinking 2.3 Applying problem solving skills 2.4 Applying observation skills
3. Convey inefficient and ineffective	3.1 Efficiency and effectiveness of resource utilization are reported to <i>appropriate personnel.</i>	3.1 Appropriate Personnel to address the	3.1 Applying written and oral communication skills

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
environmental practices	3.2 Concerns related resource utilization are discussed with appropriate personnel. 3.3 Feedback on information/ concerns raised are clarified with appropriate personnel.	environmental hazards 3.2 Environmental corrective actions	3.2 Applying critical thinking 3.3 Applying problem solving 3.4 Applying observation skills 3.5 Practicing environmental awareness

RANGE OF VARIABLES

VARIABLE	RANGE
1. Environmental Work Procedures	May include: 1.1 Utilization of Energy, Water, Fuel Procedures 1.2 Waster Segregation Procedures 1.3 Waste Disposal and Reuse Procedures 1.4 Waste Collection Procedures 1.5 Usage of Hazardous Materials Procedures 1.6 Chemical Application Procedures 1.7 Labeling Procedures
2. Appropriate Personnel	May include: 2.1 Manager 2.2 Safety Officer 2.3 EHS Offices 2.4 Supervisors 2.5 Team Leaders 2.6 Administrators 2.7 Stakeholders 2.8 Government Official 2.9 Key Personnel 2.10 Specialists 2.11 Himself

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Measured required resource utilization in the workplace using appropriate techniques 1.2 Recorded data in accordance with workplace protocol 1.3 Identified causes of inefficiency and/or ineffectiveness through deductive reasoning 1.4 Validate the identified causes of inefficiency and/or ineffectiveness thru established environmental procedures 1.5 Report efficiency and effectiveness of resource utilization to appropriate personnel 1.6 Clarify feedback on information/concerns raised with appropriate personnel
2. Resource Implications	The following resources should be provided: 2.1 Workplace 2.2 Tools, materials and equipment relevant to the tasks 2.3 PPE 2.4 Manuals and references
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration 3.2 Oral questioning 3.3 Written examination
4. Context for Assessment	4.1 Competency assessment may occur in workplace or any appropriately simulated environment 4.2 Assessment shall be observed while task are being undertaken whether individually or in-group

UNIT OF COMPETENCY : PRACTICE ENTREPRENEURIAL SKILLS IN THE WORKPLACE

UNIT CODE : 400311218

UNIT DESCRIPTOR : This unit covers the outcomes required to apply entrepreneurial workplace best practices and implement cost-effective operations.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Apply entrepreneurial workplace best practices	1.1 Good practices relating to workplace operations are observed and selected following workplace policies. 1.2 Quality procedures and practices are complied with according to workplace requirements. 1.3 Cost-conscious habits in resource utilization are applied based on industry standards.	1.1 Workplace best practices, policies and criteria 1.2 Resource utilization 1.3 Ways in fostering entrepreneurial attitudes: 1.3.1 Patience 1.3.2 Honesty 1.3.3 Quality-consciousness 1.3.4 Safety-consciousness 1.3.5 Resourcefulness	1.1 Applying communication skills 1.2 Complying with quality procedures
2. Communicate entrepreneurial workplace best practices	2.1 Observed good practices relating to workplace operations are communicated to appropriate person . 2.2 Observed quality procedures and practices are communicated to appropriate person. 2.3 Cost-conscious habits in resource utilization are communicated based on industry standards.	2.1 Workplace best practices, policies and criteria 2.2 Resource utilization 2.3 Ways in fostering entrepreneurial attitudes: 2.3.1 Patience 2.3.2 Honesty 2.3.3 Quality-consciousness 2.3.4 Safety-consciousness 2.3.5 Resourcefulness	2.1 Applying communication skills 2.2 Complying with quality procedures 2.3 Following workplace communication protocol
3. Implement cost-effective operations	3.1 Preservation and optimization of workplace resources is implemented in	3.1 Optimization of workplace resources 3.2 5S procedures and concepts	3.1 Implementing preservation and optimizing workplace resources

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	accordance with enterprise policies. 3.2 Judicious use of workplace tools, equipment and materials are observed according to manual and work requirements. 3.3 Constructive contributions to office operations are made according to enterprise requirements. 3.4 Ability to work within one's allotted time and finances is sustained.	3.3 Criteria for cost-effectiveness 3.4 Workplace productivity 3.5 Impact of entrepreneurial mindset to workplace productivity 3.6 Ways in fostering entrepreneurial attitudes: 3.6.1 Quality-consciousness 3.6.2 Safety-consciousness	3.2 Observing judicious use of workplace tools, equipment and materials 3.3 Making constructive contributions to office operations 3.4 Sustaining ability to work within allotted time and finances

RANGE OF VARIABLES

VARIABLE	RANGE
1.Good practices	May include: 1.1 Economy in use of resources 1.2 Documentation of quality practices
2.Resources utilization	May include: 2.1 Consumption/ use of consumables 2.2 Use/Maintenance of assigned equipment and furniture 2.3 Optimum use of allotted /available time

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Demonstrated ability to identify and sustain cost-effective activities in the workplace 1.2 Demonstrated ability to practice entrepreneurial knowledge, skills and attitudes in the workplace
2. Resource Implications	The following resources should be provided: 2.1 Simulated or actual workplace 2.2 Tools, materials and supplies needed to demonstrate the required tasks 2.3 References and manuals 2.3.1 Enterprise procedures manuals 2.3.2 Company quality policy
3. Methods of Assessment	Competency in this unit should be assessed through: 3.1 Interview 3.2 Third-party report
4. Context of Assessment	4.1 Competency may be assessed in workplace or in a simulated workplace setting 4.2 Assessment shall be observed while tasks are being undertaken whether individually or in-group

COMMON COMPETENCIES

UNIT OF COMPETENCY : UPDATE INDUSTRY KNOWLEDGE

UNIT CODE : SOC532201

UNIT DESCRIPTOR : This unit of competency deals with the knowledge, skills required to access, increases and update industry knowledge.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Seek and apply information on the industry	1.1 <i>Sources of information</i> on the industry are correctly identified and accessed. 1.2 <i>Information to assist effective work performance</i> is obtained in line with job requirements. 1.3 Specific information on sector of work is accessed and updated. 1.4 Industry information is correctly applied to day-to-day work activities.	COMMUNICATION 1.1 Industry information sources 1.2 Overview of quality assurance in the industry 1.3 Role of individual staff members	1.1 Ready skills needed to access industry information 1.2 Basic competency skills needed to access the internet
2. Update industry knowledge	2.1 Informal and/or formal research is used to update general knowledge of the industry. 2.2 Updated knowledge is shared with customers and colleagues as appropriate and incorporated into day-to-day working activities.	COMMUNICATION 2.1 Role of individuals in a creative endeavor member 2.2 Sources of Industry information	2.1 Time management skills 2.2 Ready skills needed to access industry information

RANGE OF VARIABLES

VARIABLE	RANGE
1. Sources of information	May include: 1.1 Media 1.2 Reference materials 1.3 Industry associations 1.3.1. Mentors 1.3.2. Training institutions 1.3.3. Technical organizations 1.4 Industry journals 1.5 Internet 1.6 Personal observation and experience
2. Information to assist effective work performance	May include: 2.1 Different sectors of the industry and the services available in each sector 2.2 Awareness on different culture 2.3 Relationship between the industry and other industries 2.4 Industry working conditions 2.5 Legislation that affects the industry 2.5.1 Dangerous Drug Act (DDA) 2.5.2 Intellectual Property Ownership (IPO) 2.5.3 Health and safety 2.5.4 Hygiene 2.5.5 Labor work practices 2.5.6 Workers' rights and compensation 2.5.7 Viewer advisory 2.5.8 Building and other related regulations 2.5.9 Other related legislations 2.6 Guilds and associations 2.7 Industrial relations issues and major organizations 2.8 Career opportunities within the industry 2.9 Work ethics 2.10 Quality assurance

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Sought and applied information on the industry 1.2 Updated industry knowledge
2. Resource Implications	The following resources should be provided: 2.1 Tools, prepress equipment/machine and facilities appropriate to processes or activity 2.2 Supplies and materials relevant to the proposed activity 2.3 Appropriate applications 2.4 Sample files
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Direct observation of activities with questioning 3.2 Oral or written questioning 3.3 Case study 3.4 Interview
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : MANAGE OWN PERFORMANCE

UNIT CODE : SOC514202

UNIT DESCRIPTOR : This unit of competency covers the knowledge, skills and attitudes in effectively managing own workload, resources and quality work.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Plan for completion of own workload	1.1 Tasks are identified according to work instructions. 1.2 Work order and schedules are designed and organized based on timelines/deadlines. 1.3 Team coordination is applied when required in completion of workload.	COMMUNICATION 1.1 Teamwork 1.2 Resource management 1.3 Timelines	1.1 Planning and organizing workload and resources 1.2 Communication skills
2. Maintain quality of performance	2.1 Personal performance is monitored according to established plant standards. 2.2 Advice and guidance are obtained when necessary to achieve or maintain established standards. 2.3 Guidance from management when necessary is applied to achieve according to established standards.	COMMUNICATION 2.1 Indicators of appropriate performance for each area of responsibility 2.2 Steps for improving or maintaining performance	2.1 Ability to observe and record performance-related concerns and information
3. Evaluate and assess own work	3.1 Actual work output is evaluated and assessed according to work instructions. 3.2 Work productivity is evaluated according to established procedures. 3.3 Feedback is obtained from relevant persons.	COMMUNICATION 3.1 Project Management 3.2 Process documentation	3.1 Managing work orders 3.2 Evaluating own performance 3.3 Reporting of daily production activities

RANGE OF VARIABLES

VARIABLE	RANGE
1. Tasks	May be identified through: 1.1 Assigned task 1.2 Verbal Instructions 1.3 Policy Documents 1.4 Project brief including timelines and schedules
2. Work order and schedules	May include: 2.1 Gantt charts 2.2 Production schedule 2.3 Delivery dates

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Planned for completion of own workload 1.2 Maintained quality of performance 1.3 Evaluated and assessed own work
2. Resource implications	The following resources should be provided: 2.1 Access to relevant venue, equipment and materials 2.2 Assignment Instructions 2.3 Logbooks 2.4 Calendar of activities 2.5 Sample liquidation and report of expenses
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/observation with oral questioning
4. Context of assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : MAINTAIN SAFE, CLEAN AND EFFICIENT WORK ENVIRONMENT

UNIT CODE : SOC514204

UNIT DESCRIPTOR : This unit of competency covers the knowledge, skills and attitudes needed to maintain clean, safe and efficient working environment. The unit incorporates the work safety guidelines.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWL EDGE	REQUIRED SKILLS
1. Comply with safety and health regulations	1.1 <i>Safety and health regulations</i> are identified and complied with. 1.2 <i>Policies and procedures</i> are adapted and applied.	ENVIRONMENT 1.1 OSHS policies and standards 1.2 Fire code	1.1 Complying with health and safety regulations 1.2 Reading and comprehension
2. Assess work area	2.1 Work areas and walkways are maintained in a safe and hazard free environment. 2.2 All routines are carried out in accordance with Occupational Safety and Health Standards (OSHS). 2.3 Waste is stored and disposed of according to Occupational Safety and Health Standards (OSHS).	ENVIRONMENT 2.1 Work Hazards Policies and Procedures 2.2 OSHS policies and procedures 2.3 Waste management	2.1 Complying with health and safety regulations
3. Check and maintain tools, equipment and resources	3.1 Tools, equipment and resources are stored according to safety regulations. 3.2 Tools, equipment and resources are checked for maintenance requirements. 3.3 Tools and equipment are referred for repair as required.	ENVIRONMENT 3.1 Maintenance of tools and equipment 3.2 Tools, equipment and resource maintenance requirements	3.1 Checking for maintenance requirements 3.2 Storing tools and equipment

RANGE OF VARIABLES

VARIABLE	RANGE
1. Safety and Health Regulations	May include: 1.1 Clean Air Act 1.2 National Building Code 1.3 Philippine Electrical Code 1.4 Fire Code of the Philippines 1.5 Waste management statutes and rules 1.6 Philippine Occupational Safety and Health Standards 1.7 DOLE OSH related issuances 1.8 ECC regulations
2. Policies and procedures	May include: 2.1 Hazard Policies and Procedures 2.2 Emergency, Fire and Accident Procedures 2.3 Personal Safety Procedures 2.4 Procedures for the use of personal protective clothing and equipment 2.5 Hazard Identification 2.6 Standard Operating Procedures

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Complied with safety and health regulations 1.2 Assessed work area 1.3 Checked and maintained tools, equipment and resources
2. Resource implications	The following resources should be provided: 2.1 Access to relevant venue, tools, equipment and resources to perform the tasks 2.2 Required operating manual/s 2.3 Safety regulations 2.4 Relevant policies and procedures
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/Observation with oral questioning
4. Context of assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM BASIC COMPUTER OPERATIONS

UNIT CODE : ELC311203

UNIT DESCRIPTOR : This unit covers the knowledge, skills, attitudes and values needed to identify, select, and perform basic operations and maintenance of a desktop publishing system.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify the basic components of a desktop publishing system	1.1 Hardware of a computer are identified according to desktop publishing system requirements. 1.2 Software is identified according to desktop publishing system requirements. 1.3 Other peripherals are identified according to desktop publishing system requirements.	TECHNOLOGY 1.1 Basic components of a computer system 1.2 Computer specifications for desktop color publishing system 1.3 Various operating systems and platforms	1.1 Identifying kinds of computer components 1.2 Identifying computer software 1.3 Identifying computer peripherals
2. Select the appropriate hardware, software and other peripherals needed for desktop publishing	2.1 Appropriate hardware and its specifications needed for desktop publishing are selected. 2.2 Appropriate software needed for desktop publishing is selected. 2.3 Peripherals needed for desktop publishing is selected.	TECHNOLOGY 2.1 Appropriate hardware for desktop publishing 2.2 Appropriate software for desktop publishing	2.1 Selecting appropriate hardware and software 2.2 Selecting appropriate peripherals for desktop publishing
3. Operate the basic components of a desktop publishing system	3.1 Files in the server is accessed via network . 3.2 Appropriate software is used according to work instructions. 3.3 Other peripherals are used properly according to manufacturer's recommendation.	TECHNOLOGY 3.1 Desktop taskbars and icons 3.2 Local drive, partition drive and removable drive functions 3.3 Address bar and internet browsing 3.4 Networked computer connection	3.1 Accessing files 3.2 Operating appropriate software 3.3 Written and oral communications 3.4 Following manufacturer's recommendation
4. Maintain the computer system and	4.1 Computer is checked regularly for any virus	TECHNOLOGY 4.1 Various types of virus and its	4.1 Checking of virus and malicious software

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
its external parts	and malicious software. 4.2 External parts of the computer are cleaned regularly. 4.3 Computer desktop is de-cluttered in accordance with file management procedures.	effects on the computer 4.2 Various types of malicious software and its effects on the computer 4.3 File management system ENVIRONMENT 4.4 Proper computer cleaning	4.2 Removing virus that affects the system 4.3 Recovering files affected by malicious software 4.4 Cleaning the external parts of the computer

RANGE OF VARIABLES

VARIABLE	RANGE
1. Hardware	May include: 1.1 The central processing unit (CPU) 1.2 Monitor 1.3 Keyboard and mouse 1.4 Output devices
2. Software	May include 2.1 Operating Systems 2.2 Software application 2.3 Utility program
3. Other Peripherals	May include: 3.1 Microphone 3.2 Speaker 3.3 Headset 3.4 External drives 3.5 Scanner 3.6 Barcode/Quick Response (QR) code reader 3.7 Internet modem 3.8 Pen tool 3.9 Network hub
4. Appropriate Hardware	May include: 4.1 Extended keyboard 4.2 Required computer for workstation 4.3 High-end monitor 4.4 Postscript (PS) or Printer Command Language (PCL) driven printer
5. Specifications	May include: 5.1 Required capacity and speed of Random Access Memory (RAM) 5.2 Processor clock speed and processing capacity 5.3 Hard Drive / Solid-state Drive speed and capacity 5.4 Video card speed and capacity
6. Appropriate software application	May include: 6.1 Adobe InDesign – for design and layout 6.2 Adobe Illustrator – for illustration and drawing 6.3 Adobe Photoshop – for image editing and manipulation 6.4 Microsoft Word – for typesetting 6.5 Microsoft Excel – for variable data encoding 6.6 Portable Document Format – For viewing/review 6.7 Scanning software – for scanning
7. Network	May include: 7.1 Personal Area Network (PAN) 7.2 Local Area Network (LAN)

VARIABLE	RANGE
	7.3 Wireless Local Area Network (WLAN) 7.4 Remote or Virtual Private Network (VPN)
8. Virus and malicious software	May include: 8.1 Trojan 8.2 Worms 8.3 Hybrids and exotic forms 8.4 Ransomware 8.5 Fileless malware 8.6 Adware 8.7 Malvertising 8.8 Spyware
9. External parts	May include: 9.1 Keyboard 9.2 Monitor 9.3 Mousepad 9.4 CPU Case 9.5 Computer fan

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified the basic components of a desktop publishing system 1.2 Selected the appropriate hardware, software and other peripherals needed for desktop publishing 1.3 Operated the basic components of a desktop publishing system 1.4 Maintained the computer system and its external parts
2. Resource Implications	The following resources should be provided: 2.1 Tools, prepress equipment/machines and facilities appropriate to processes or activity 2.2 Supplies and materials relevant to the proposed activity 2.3 Appropriate applications 2.4 Sample files
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Direct observation of activities with questioning 3.2 Oral or written questioning 3.3 Case study 3.4 Interview
4. Context of Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

CORE COMPETENCIES

UNIT OF COMPETENCY : PERFORM HALFTONE REPRODUCTION AND ITS REQUIREMENTS

UNIT CODE : SOC732306

UNIT DESCRIPTOR : This unit covers the knowledge skills and attitudes required to perform halftone reproduction and its requirements and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify halftone reproduction and its requirements	1.1 File copy is identified for halftone reproduction. 1.2 Image capturing devices are identified & prepared according to the task to be undertaken. 1.3 Appropriate software is selected in accordance to specified reproduction requirements.	TECHNOLOGY 1.1 Image capturing devices preparation 1.2 Selection of appropriate software 1.3 Computer operation COMMUNICATION 1.4 Identification of halftone reproduction 1.5 Verbal and written communication	1.1 Effective communication skills 1.2 Determining differences between hard and soft copy 1.3 Operating different image capturing devices 1.4 Applying scanner software 1.5 6S principles
2. Convert a continuous tone copy into a printable image	2.1 Materials to be reproduced are checked if complete, fully identified & had complied according to work instructions. 2.2 Checking is conducted on original copy to be reproduced. 2.3 Image capturing device settings are set according to established procedures. 2.4 Hard copy is captured in accordance to specified reproduction requirements .	COMMUNICATION 2.1 Job specification 2.2 Client requirements 2.3 Flow of communication 2.4 Task planning and preparation 2.5 Information on characteristic of materials	2.1 Gathering client's specifications 2.2 Checking client's requirements 2.3 Handling / keeping supplied materials with care 2.4 Planning and preparing task

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Recommend appropriate halftone dot frequency and dot shapes	3.1 <i>Paper-ink-press characteristics</i> is determined according to work instructions. 3.2 Appropriate <i>halftone specifications</i> are recommended in accordance with established procedures. 3.3 Copy of reproduced image is saved in an appropriate format for the next printing process.	TECHNOLOGY 3.1 Computer functions 3.2 Various software applications and their usage 3.3 Installation of identified various types of fonts	3.1 Computer skills 3.2 Using different software applications related to printing 3.3 Identifying and installing various fonts

RANGE OF VARIABLES

VARIABLE	RANGE
1. Copy	May include: 1.1 Hard Copy 1.2 Soft Copy
2. Image Capturing Devices	May include: 2.1 Scanner 2.2 Digital Camera 2.3 Video Camera 2.4 Mobile Camera 2.5 3D Camera 2.6 Print Screen
3. Appropriate Software Application	May include: 3.1 Scanning Software 3.2 Image Manipulation and Editing Software
4. Materials	May include: 4.1 Reflection Copy 4.2 Transparencies 4.3 Artwork Materials
5. Check on original copy	May include: 5.1 Color Quality 5.2 Sharpness 5.3 Contrast
6. Device Settings	May include: 6.1 Image resolution 6.2 Magnification 6.3 Color Mode 6.4 Sharpness Settings
7. Specified Reproduction Requirements	May include: 7.1 Tonal Correction 7.2 Color Correction 7.3 Detail Enhancement
8. Paper-ink-press Characteristics	May include: 8.1 Substrate 8.2 Ink 8.3 Press
9. Halftone Specifications	May include: 9.1 Frequency (AM, FM, Hybrid) 9.2 Dot Shapes

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified file copy for halftone reproduction. 1.2 Identified and prepared image capturing devices. 1.3 Selected appropriate software. 1.4 Checked completeness, fully identified and complied materials to be reproduced. 1.5 Conducted checking on original copy to be reproduced. 1.6 Set image capturing device settings. 1.7 Captured hard copy according to specified reproduction requirements. 1.8 Determined paper-ink-press characteristics. 1.9 Recommended appropriate halftone specifications.
2. Resource Implications	The following resources should be provided: 2.1 Workshop appropriate for the unit of competency 2.2 Tools, materials and equipment, appropriate for the unit of competency 2.3 Video clips 2.4 Case work
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with questioning 3.2 Observation with questioning 3.3 Written Examination 3.4 Interview
4. Context for Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM BASIC COLOR SEPARATION PROCESS

UNIT CODE : SOC732307

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes needed to prepare and perform basic color separation process as well as saving of an image. It includes manual coloring of a drawing, text fill/outline or background and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Check requirements for basic color separation process	1.1 Basic color separation requirements are determined according to work instructions. 1.2 Drawing source is determined according to work instructions. 1.3 Requirements for basic color separation of a drawing is checked according to work instructions.	TECHNOLOGY 1.1 Color Theory 1.2 Calibration of paper-ink-press system 1.3 Types of basic color separation 1.4 Types of images 1.5 Types of screen tint	1.1 Checking work instructions 1.2 Drawing skills
2. Prepare hardware and software necessary for basic color separation	2.1 Digitization of artwork materials is performed using image capturing device in accordance with established procedures. 2.2 Software is set in preparation for basic color separation process. 2.3 Necessary colors needed for basic color separation are prepared in accordance with work instructions.	TECHNOLOGY 2.1 Various color separation software application 2.2 Various image capturing device operation 2.3 Color/spot color preparation	2.1 Digitizing artwork materials 2.2 Applying color separation software 2.3 Computer skills
3. Perform basic color separation of an image	3.1 Layers are created according to established procedures. 3.2 Areas to be applied with colors are selected using the selection tools . 3.3 Color screen tint, gradation and pattern are applied into the selected	TECHNOLOGY 3.1 Layer creation 3.2 Archiving procedures 3.3 Selection and coloring tools COMMUNICATION	3.1 Layering of digital file 3.2 Applying selection and coloring tools 3.3 Archiving a copy 3.4 Drawing/tracing techniques

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	areas using coloring tools . 3.4 Copy for archiving purposes is saved in accordance with established procedures.	3.4 Concept of color separation	
4. Print a copy for checking	4.1 A colored proof is printed for checking in accordance with established procedures. 4.2 Printed copy is checked based on the original copy and work instructions. 4.3 Printed copy is forwarded to relevant person for approval. 4.4 Correction and revision are applied to the proof submitted by relevant person.	COMMUNICATION 4.1 Checking of printed copy against work instruction 4.2 Application of correction and revision	4.1 Printing of proof 4.2 Checking printed copy 4.3 Correcting and revising skills 4.4 Communication skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Requirements	May include: 1.1 Number of color combination 1.2 Types of screen application 1.3 Types of copy to be separated
2. Drawing Source	May include: 2.1 Artwork Material 2.2 Digital Artwork 2.3 Internet Downloaded Artwork
3. Checked	May include: 3.1 Image Quality 3.2 Number of Colors 3.3 Software Applications to be Used 3.4 Types of Color Application 3.5 Types of Screen 3.6 Types of Copy to be Separated
4. Digitization	May include: 4.1 Through Scanning 4.2 Through Digital Camera
5. Software	May include: 5.1 Vector Graphic Editor 5.2 Image Manipulation and Editing 5.3 Design and Lay-out
6. Prepared	May include: 6.1 Create color in the color swatches 6.2 Create additional channels for spot color 6.3 Substitute spot color with any of the Cyan, Magenta, Yellow, Black colors
7. Selection Tools	May include: 7.1 Magic Wand 7.2 Pen Tool 7.3 Selection Tools
8. Coloring Tools	May include: 8.1 Brush Tool 8.2 Pencil Tool 8.3 Paint Bucket Tool 8.4 Gradient Tool 8.5 Blend Tool 8.6 Mesh Tool 8.7 Mixer Brush Tool 8.8 Color Replacement Tool
9. Archiving	May include:

VARIABLE	RANGE
	9.1 Vectored Graphic 9.2 Rastered Graphic File 9.3 Tagged Image Format File 9.4 Design and Lay-out File 9.5 Portable Document Format file 9.6 Encapsulated Postscript file
10. Colored Proof	May include: 10.1 Inkjet Printed 10.2 Colored Laser Printed
11. Checked	May include: 11.1 Correct Size 11.2 Correct Color 11.3 Sharpness

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Checked requirements for basic color separation of a drawing according to work instructions 1.2 Performed digitization of artwork materials using image capturing device 1.3 Prepared necessary colors needed for basic color separation in accordance with work instructions 1.4 Applied color screen tint, gradation and pattern into the selected areas using coloring tools 1.5 Evaluated printed copy based on the original copy and work instructions 1.6 Applied correction and revision to the proof submitted by relevant person
2. Resource Implications	The following resources should be provided: 2.1 Workshop appropriate for the unit of competency 2.2 Tools, materials and equipment, appropriate for the unit of competency 2.3 Video Clips 2.4 Case Work
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with questioning 3.2 Observation with questioning 3.3 Written Examination 3.4 Interview
4. Context for Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM DIGITAL OUTPUT PROCESSING OPERATIONS

UNIT CODE : SOC732308

UNIT DESCRIPTOR : This unit covers the knowledge skills and attitudes required to perform digital output process which includes proofing, plate recording, die-knife guide making operations, and the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Perform digital contract proofing operations	1.1 Digital printer is checked if ready and in accordance with established procedures. 1.2 Printing requirements of a job is prepared according to work instructions. 1.3 Data file is printed according to established procedures. 1.4 Printed proof is quality checked according to established procedures. 1.5 Checked proof is forwarded to relevant personnel in accordance with established procedures.	TECHNOLOGY 1.1 Functions of various proofing machines 1.2 Various types of proof 1.3 File handling 1.4 Preflight procedures COMMUNICATION 1.5 Quality control procedures	1.1 Understanding work instructions 1.2 Preparing printing requirements 1.3 Checking of files 1.4 Checking of printing quality 1.5 Oral and written communications
2. Perform plate recording and processing methods	2.1 Computer-to-Plate (CtP) system is selected and according to work instructions. 2.2 Computer-to-Plate system is prepared according to established procedure. 2.3 Ready-file is reviewed before sending to Raster Image Processor (RIP). 2.4 Plate making make-ready is performed according to established procedures. 2.5 Ripped file is examined against the contract	TECHNOLOGY 2.1 Parts and functions of Computer-to-Plate equipment 2.2 Make-ready of plate making 2.3 Raster Image Processor functions 2.4 Image Recording System COMMUNICATION 2.5 Plate quality inspections ENVIRONMENT	2.1 Preparing computer to plate system 2.2 Ripping of data files 2.3 Checking of proof 2.4 6's principles

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	proof according to established procedures. 2.6 <i>Imaging procedures</i> are performed according to established procedures. 2.7 Plate is <i>inspected</i> and forwarded to relevant persons for the next printing process.	2.6 Proper handling of consumables	
3. Perform die-cutting guide for die-knife making output operations	3.1 <i>Source</i> of die-cutting guide dimensions are determined based on work instructions. 3.2. Die-cutting guide making is <i>performed</i> using vector graphic editor software according to established procedures. 3.2 <i>Line assignments</i> are set according to work instructions. 3.3 Finished die-cutting guide layout is saved in the <i>file extension format/s</i>. 3.4 Finished die-cutting guide layout is printed for <i>inspections</i>. 3.5 <i>Mocked-up process</i> is performed according to work instructions. 3.6 Mock-up sample is presented to <i>relevant person</i> for approval.	TECHNOLOGY 3.1 Vector graphic editor software MATHEMATICS 3.2 Mensuration ENVIRONMENT 3.3 Mock-process procedures	3.1 Measuring skills 3.2 Drawing skills 3.3 Managing data 3.4 Printing of layout 3.5 Forming skills 3.6 Communication skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Ready	May include: 1.1 Power on 1.2 Printer is online 1.3 Nozzle tested and cleaned 1.4 Enough ink contents 1.5 Enough media 1.6 Raster Image Processor (RIP) is ready
2. Printing Requirements	May include: 2.1 Correct media 2.2 Color profile assignment
3. Data File	May include: 3.1 Latest approved in-house produced data file 3.2 Archived file for repeat orders 3.3 Client's supplied ready-to-print file
4. Quality Checked	May include: 4.1 No banding 4.2 Actual size 4.3 No font errors 4.4 Standard image quality 4.5 All elements are present 4.6 No unwanted images 4.7 Correct color profile assignment
5. Prepared	May include: 5.1 Recording unit is ready 5.2 Developing and stacking units are ready 5.3 Raster image processing computer is ready 5.4 Raster image processing software is on-line 5.5 Plate calibration is set according to standards 5.6 Required room lights are ready
6. Reviewed	May include: 6.1 All required marks are present 6.2 All required marks are in the correct placement 6.3 All elements of a ready-file are present compared to approved contract proof 6.4 Ready-file layout size is correct based on computer-to-plate output instructions
7. Make-ready	May include: 7.1 Converting of data file into Portable Document File (.pdf) or Postscript file (.ps) 7.2 Setting of halftone frequency 7.3 Setting of dot shapes 7.4 Setting of plate gripper margin

VARIABLE	RANGE
	7.5 Importing converted data into the Raster Image Processor for 1-bit processing 7.6 Checking of Raster Image Processor logs for errors
8. Examined	May include: 8.1 All fonts are present and correct 8.2 All layout elements are present 8.3 Image trapping and overprint are well performed 8.4 Correct special color and process color assignments 8.5 Single color text and lines are properly performed 8.6 Correct dot shapes 8.7 No unwanted images
9. Imaging Procedures	May include: 9.1 Plate recording 9.2 Plate developing/washing 9.3 Plate gumming 9.4 Plate drying 9.5 Plate stacking
10. Inspected	May include: 10.1 Plate dent 10.2 Plate scratches 10.3 Skewed image 10.4 Wrong gripper margins 10.5 Unwanted images 10.6 All elements are present according to contract proof 10.7 Plate identifications are properly labelled 10.8 No broken text or images 10.9 Plate control wedge did not meet established standards
11. Source	May include: 11.1 Client's supplied dimensions 11.2 Sample copy dimensions 11.3 Dimensions based on work instructions for development stage
12. Performed	May include: 12.1 Drawing of straight lines and corners 12.2 Drawing of curves lines 12.3 Setting the offset of lines and angles
13. Line assignments	May include: 13.1 Assigning lines for through/kiss blades 13.2 Assigning lines for scoring/creasing blades 13.3 Assigning lines for perforating blades
14. File Extension Format/s	May include: 14.1 Vectored Graphic Format 14.2 Portable Document Format

VARIABLE	RANGE
	14.3 Encapsulated Postscript
15. Inspections	May include: 15.1 Correct layout size 15.2 Correct line assignments
16. Mocked-up process	May include: 16.1 Cutting 16.2 Folding 16.3 Gluing
17. Relevant Person	May include: 17.1 Client 17.2 Account Executive 17.3 Customer Service Representative 17.4 Production Manager 17.5 Prepress Supervisor 17.6 Team Leader 17.7 Designer

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared printing requirements of a job 1.2 Performed quality check of printed proof 1.3 Reviewed ready-file before sending to Raster Image Processor (RIP) 1.4 Performed plate making make-ready 1.5 Examined ripped file against the contract proof 1.6 Inspected plate and submitted to relevant persons for the next printing process 1.7 Performed die-cutting guide making using vector graphic editor software 1.8 Set line assignments
2. Resource Implications	The following resources should be provided: 2.1 Workshop appropriate for the unit of competency 2.2 Tools, materials and equipment, appropriate for the unit of competency 2.3 Video clips 2.4 Case work
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with questioning 3.2 Observation with questioning 3.3 Written Examination 3.4 Interview
4. Context for Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : PERFORM IMPOSITION PROCEDURES

UNIT CODE : SOC732309

UNIT DESCRIPTOR : This unit covers the knowledge, skills, attitudes, and values required to perform imposition procedures which include creating, nesting, inserting, imposing, and reviewing data using the appropriate materials and software. It also includes the application of commonly used tools and equipment.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Create a master layout for publication	1.1 Pagination of folding dummy is checked against soft copy pagination. 1.2 Master layout is created using the appropriate software in accordance with work instructions. 1.3 Imposition measurements are set in accordance with established procedures.	TECHNOLOGY 1.1 Computer application MATHEMATICS 1.2 Basic measurement COMMUNICATION 1.3 Verbal and written communication 1.4 Various types of folding dummy 1.5 Press and post-press requirements 1.6 Concept of imposition	1.1 Communication Skills 1.2 Interpersonal skills 1.3 Operating computer 1.1 Performing imposition application
2. Perform nesting of jobs	2.1 Design layout and die-cutting layout are checked if all elements are complete according to work instructions. 2.2 Die-cutting layout is checked if it conforms with the size of the design layout. 2.3 Layout with die-cutting guide is compared against the mock-up sample provided by the client. 2.4 Layout with or without die-cutting guide is	COMMUNICATION 2.1 Procedures on producing packaging products 2.2 Concept of job nesting 2.3 Die-cutting knife function 2.4 Die-line interpretation ENVIRONMENT 2.5 Quality control methods	2.1 Performing quality inspection 2.2 Nesting of jobs 2.3 Interpreting work instructions 2.4 Applying line reproduction requirements

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	nested in accordance with work instructions. 2.5 Nested jobs are checked to avoid printing errors according to established quality control procedures.		
3. Insert required marks for press and post-press	3.1 Required marks for press are inserted according to press established procedures. 3.2 Required marks for post-press are inserted according to post-press established procedures. 3.3 Other information marks are inserted according to established procedures.	COMMUNICATION 3.1 Functions of marks 3.2 Marks for press requirements 3.3 Marks of post-press requirements 3.4 Information marks needed for archiving	3.1 Inserting various types of marks 3.2 Positioning of marks
4. Perform manual / automatic imposition	4.1 Individual pages are imposed either manually or electronically. 4.2 Imposition is reviewed according to established company standards. 4.3 Final imposed job is saved in preparation for the next process.	COMMUNICATION 4.1 Concept of manual imposition 4.2 Concept of electronic imposition 4.3 Archiving procedures	4.1 Imposing manually 4.2 Imposing electronically 4.3 Reviewing imposed jobs
5. Print a small scale sample for checking	5.1 Small scale sample (25%) of the imposed job is printed for final imposition check. 5.2 Folding sequence is checked in accordance with the binding requirements. 5.3 Folded pages are arranged in order in accordance with work instruction.	TECHNOLOGY 5.1 Advance printer's preference COMMUNICATION 5.2 Requirements for binding 5.3 Requirements for nesting	5.1 Collating signatures 5.2 Insertions of signatures 5.3 Checking of printed copies

RANGE OF VARIABLES

VARIABLE	RANGE
1. Folding Dummy	May include: 1.1 For Web-press folding 1.2 For stand-alone folding machine 1.3 For saddle stitching 1.4 For smyth sewn binding 1.5 For special type of folding
2. Soft Copy	May include: 2.1 Design and Lay-out 2.2 Vectored Graphic 2.3 Encapsulated Postscript 2.4 Portable Document Format 2.5 Rastered Graphic
3. Master Layout	May include: 3.1 Books master layout 3.2 Magazine master layout 3.3 Journal master layout 3.4 Brochure master layout
4. Imposition Measurement	May include: 4.1 Head distance 4.2 Creep 4.3 Folding allowance 4.4 Spine allowance 4.5 Trimming allowance 4.6 Final trim size 4.7 Gripper allowance
5. Layout	May include: 5.1 Approved design 5.2 Client supplied file 5.3 Repeat order job
6. Die-cutting	May include: 6.1 Creasing blade 6.2 Perforation blade 6.3 Cutting blade
7. Mock-up sample	May include: 7.1 Sample product 7.2 Prototype 7.3 Contract proof
8. Nested	May include: 8.1 Single job step and repeat 8.2 Multiple design step and repeat
9. Printing Errors	May include: 9.1 Unwanted image cropping

VARIABLE	RANGE
	9.2 Wrong orientation of image against media grain direction 9.3 Measurement errors 9.4 Folding errors
10. Marks for Press	May include: 10.1 Registration marks 10.2 Print control bars 10.3 Drop guide marks 10.4 Side guide marks 10.5 Print color indicator 10.6 Plate marks 10.7 Plate control wedge
11. Marks for post press	May include: 11.1 Folding guide marks 11.2 Trimming marks 11.3 Bleed marks 11.4 Collating index marks 11.5 Quality inspection marks 11.6 Gluing Marks
12. Information Marks	May include: 12.1 Security marks 12.2 Job information box
13. Reviewed	May include: 13.1 Duplication of pages 13.2 Missing pages 13.3 Alignment of imposed pages 13.4 Overlapped pages 13.5 Inadequate/excess bleeds 13.6 Total number of flats 13.7 Cropped pages 13.8 Page rotation 13.9 Presence of marks 13.10 Margins
14. Final imposition check	May include: 14.1 Correct order of pages 14.2 Correct rotation of pages 14.3 Correct order of signatures 14.4 Required marks
15. Binding requirements	May include: 15.1 Single perfect bind 15.2 Double roll perfect bind 15.3 Saddle bind 15.4 Smyth bind 15.5 Side stitch bind

VARIABLE	RANGE
	15.6 Other type of binding
16.Order	May include: 16.1 Insertion gathering 16.2 Collating gathering

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Created master layout using the appropriate software 1.2 Set imposition measurements 1.3 Checked design layout and die-cutting layout if all elements are complete 1.4 Checked die-cutting layout if it conforms with the size of the design layout 1.5 Nested layout with or without die-cutting guide 1.6 Checked nested jobs to avoid printing errors 1.7 Inserted required marks for press 1.8 Inserted required marks for post-press 1.9 Imposed individual pages either manually or electronically 1.10 Reviewed imposition according to established company standards 1.11 Checked folding sequence in accordance with the binding requirements
2. Resource Implications	The following resources should be provided: 2.1 Workshop appropriate for the unit of competency 2.2 Tools, materials and equipment, appropriate for the unit of competency
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with questioning 3.2 Observation with questioning 3.3 Written Examination 3.4 Interview
4. Context for Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

**UNIT OF COMPETENCY : PERFORM REGULAR MAINTENANCE OF
CONDITION-BASED PREPRESS
MACHINES/EQUIPMENT**

UNIT CODE : SOC532310

UNIT DESCRIPTOR : This unit deals with the knowledge, skills and discipline needed to monitor the actual condition of prepress machines/equipment and recommend what maintenance needed to be done.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Perform inspection on major parts of equipment/machines based on the agreement with the supplier	1.1 Inspection is performed regarding the wear and tear of major parts of equipment/machines. 1.2 Inspections are performed on ink/water/air flow systems of equipment/machines for possible clogging and damage. 1.3 Inspections are performed on the output quality of various digital outputting machines.	TECHNOLOGY 1.1 Parts of equipment/machines 1.2 Equipment/machines consumables 1.3 Ink/water/air flow systems 1.4 Output quality standards 1.5 Various digital prepress machines COMMUNICATION 1.6 Supplier-client maintenance agreement	1.1 Inspecting parts of prepress machines/equipment 1.2 Inspecting output quality 1.3 Inspecting ink/water/air flow systems
2. Perform fault finding check on the inspected parts of equipment/machines	2.1 Fault finding check is performed on machine parts in accordance with established procedures. 2.2 Fault finding check is recorded in accordance with established procedures. 2.3 Fault finding check reports are submitted to relevant person for further actions.	COMMUNICATION 2.1 Fault finding check on various prepress machines/equipment	2.1 Communication skills 2.2 Analytical skills 2.3 Critical thinking skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Major parts	May include: 1.1 Printhead 1.2 Rollers 1.3 Cutting plotter blade 1.4 Machine power cords and other external cords 1.5 Diode
2. Ink/water/air flow systems	May include: 2.1 Printer's ink tubing system 2.2 Plate processor's hose 2.3 Plate processor's water filter 2.4 Air-conditioning unit filter, evaporator and condenser
3. Output quality	May include: 3.1 Plate calibrations 3.2 Plate control wedge standards 3.3 Proofing machine print quality 3.4 Cutting plotter blade sharpness
4 Fault Finding Check	May include: 4.1 Machine 4.2 Process 4.3 Human

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Performed inspection regarding the wear and tear of major parts of equipment/machines. 1.2 Performed inspection on the output quality of various digital outputting machines. 1.3 Performed fault finding check on machine parts. 1.4 Recorded fault finding check. 1.5 Submitted fault finding check reports to relevant person for further actions.
2. Resource Implications	The following resources should be provided: 2.1 Workshop appropriate for the unit of competency 2.2 Tools, materials and equipment, appropriate for the unit of competency 2.3 Video clips 2.4 Case work
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration with questioning 3.2 Observation with questioning 3.3 Written Examination 3.4 Interview
4. Context for Assessment	4.1 Competency may be assessed in the actual workplace or at the designated TESDA Accredited Assessment Center.

SECTION 3 TRAINING ARRANGEMENTS

These standards are set to provide technical and vocational education and training (TVET) providers with information and other important requirements to consider when designing training programs for **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II**.

3.1 CURRICULUM DESIGN

TESDA shall provide the training on the development of competency-based curricula to enable training providers develop their own curricula with the components mentioned below.

Delivery of knowledge requirements for the basic, common and core units of competency specifically in the areas of mathematics, science/technology, communication/language and other academic subjects shall be contextualized. To this end, TVET providers shall develop a Contextual Learning Matrix (CLM) to accompany the curricula.

**Course Title: PRINTING SERVICES
(PREPRESS TECHNICAL OPERATIONS)**

NC Level: NC II

Nominal Training Duration:

37	hours	Basic Competencies
45	hours	Common Competencies
200	hours	Core Competencies
282	hours	TOTAL

Course Description:

This course is designed to provide the learner with knowledge, skills and attitude, applicable in performing work activities involve in performing halftone reproduction and its requirements, performing basic color separation process, performing digital output processing operations, performing imposition procedures and performing regular maintenance of condition-based prepress machines/equipment

Upon completion of the course, the learners are expected to demonstrate the above-mentioned competencies to be employed. To obtain this, all units prescribed for this qualification must be achieved.

BASIC COMPETENCIES (37 HOURS)

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Participate in workplace communication	1.1. Obtain and convey workplace information	- Describe organizational policies - Read: - Effective communication - Written communication - Communication procedures and systems - Identify: - Different modes of communication - Medium of communication - Flow of communication - Available technology relevant to the enterprise and the individual's work responsibilities - Prepare different types of question - Gather different sources of information - Apply storage system in establishing workplace information - Demonstrate Telephone courtesy	- Group discussion - Lecture - Demonstration	- Oral evaluation - Written examination - Observation	2 hours
	1.2. Perform duties following workplace instructions	- Read: - Written notices and instructions - Workplace interactions and procedures - Read instructions on work related forms/documents - Perform workplace duties scenario following workplace instructions	- Group discussion - Lecture - Demonstration	- Oral evaluation - Written examination - Observation	2 hours
	1.3. Complete relevant work related documents	- Describe communication procedures and systems - Read: - Meeting protocols - Nature of workplace meetings	- Group discussion - Lecture - Demonstration - Role play	- Oral evaluation - Written examination - Observation	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Workplace interactions ○ Barriers of communication • Read instructions on work related forms/documents • Practice: ○ Estimate, calculate and record routine workplace measures ○ Basic mathematical processes of addition, subtraction, division and multiplication • Demonstrate office activities in: ○ workplace meetings and discussions scenario • Perform workplace duties scenario following simple written notices • Follow simple spoken language • Identify the different Non-verbal communication • Demonstrate ability to relate to people of social range in the workplace • Gather and provide information in response to workplace requirements • Complete work-related documents			
2. Work in a team environment	2.1 Describe team role and scope	• Discussion on team roles and scope • Participate in the discussion: ○ Definition of Team ○ Difference between team and group ○ Objectives and goals of team • Locate needed information from the different sources of information	• Lecture/ Discussion • Group Work • Individual Work • Role Play	• Role Play • Case Study • Written Test	1 hour
	2.2 Identify one's role and responsibility within team	• Role play: ○ individual role and responsibility • Role Play ○ Understanding Individual differences • Discussion on gender sensitivity	• Role Play • Lecture/ Discussion	• Role Play • Written Test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.3 Work as a team member	- Participate in group planning activities - Role play : Communication protocols - Participate in the discussion of standard work procedures and practices	- Group work - Role Play - Lecture/ Discussion	- Role Play - Written Test	1 hour
3. Solve/address routine problems	3.1 Identify routine problems	- Review of the current industry hardware and software products and services - Identify correctly the industry maintenance, service and helpdesk practices, processes and procedures - Make use of the industry standard diagnostic tools - Share best practices in determining basic malfunctions and resolutions to general problems in the workplace - Analyze routine/procedural problems	- Group discussion - Lecture - Demonstration - Role playing	- Case Formulation - Life Narrative Inquiry (Interview) - Standardized test	1 hour
	3.2 Look for solutions to routine problems	- Review of the current industry hardware and software products and services - Identify correctly the industry maintenance, service and helpdesk practices, processes and procedures - Make use of the industry standard diagnostic tools - Share best practices in determining basic malfunctions and resolutions to general problems in the workplace - Formulate possible solutions to problems and document procedures for reporting	- Group discussion - Lecture - Demonstration - Role playing	- Case Formulation - Life Narrative Inquiry (Interview) - Standardized test	1 hour
	3.3 Look for solutions to routine problems	- Review of the current industry hardware and software products and services - Identify correctly the industry maintenance, service and helpdesk practices, processes and procedures - Make use of the industry standard diagnostic tools - Share best practices in determining basic malfunctions and resolutions to general problems in the workplace	- Group discussion - Lecture - Demonstration - Role playing	- Case Formulation - Life Narrative Inquiry (Interview) - Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		Formulate possible solutions to problems and document procedures for reporting			
4. Develop Career and Life Decisions	4.1 Manage one's emotion	- Demonstrate self-management strategies that assist in regulating behavior and achieving personal and learning goals - Explain enablers and barriers in achieving personal and career goals - Identify techniques in handling negative emotions and unpleasant situation in the workplace such as frustration, anger, worry, anxiety, etc. - Manage properly one's emotions and recognize situations that cannot be changed and accept them and remain professional - Recall instances that demonstrate self-discipline, working independently and showing initiative to achieve personal and career goals - Share experiences that show confidence, and resilience in the face of setbacks and frustrations and other negative emotions and unpleasant situations in the workplace	- Discussion - Interactive Lecture - Brainstorming - Demonstration - Role-playing	- Demonstration or simulation with oral questioning - Case problems involving workplace diversity issues	1 hour
	4.2 Develop reflective practice	- Enumerate strategies to improve one's attitude in the workplace - Explain Gibbs' Reflective Cycle/Model (Description, Feelings, Evaluation, Analysis, Conclusion, and Action plan) - Use basic SWOT analysis as self-assessment strategy - Develop reflective practice through realization of limitations, likes/ dislikes; through showing of self-confidence - Demonstrate self-acceptance and being able to accept challenges	- Small Group Discussion - Interactive Lecture - Brainstorming - Demonstration 5 Role-playing	- Demonstration or simulation with oral questioning - Case problems involving workplace diversity issues	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	4.3 Boost self-confidence and develop self-regulation	- Describe the components of self-regulation based on Self-Regulation Theory (SRT) - Explain personality development concepts - Cite self-help concepts (e. g., 7 Habits by Stephen Covey, transactional analysis, psycho-spiritual concepts) - Perform effective communication skills – reading, writing, conversing skills - Show affective skills – flexibility, adaptability, etc. - Determine strengths and weaknesses	- Small Group Discussion - Interactive Lecture - Brainstorming - Demonstration - Role-playing	- Demonstration or simulation with oral questioning - Case problems involving workplace diversity issues	1 hour
5. Contribute to workplace innovation	5.1 Identify opportunities to do things better	- Identify different roles of individuals in contributing to doing things better in the workplace - Explain the concepts of positive impacts and challenges in innovation - Show mastery of the different types of changes and levels of participation in the workplace - Discuss 7 habits of highly effective people	- Interactive Lecture - Appreciative Inquiry - Demonstration - Group work	- Psychological and behavioral Interviews - Performance Evaluation - Life Narrative Inquiry - Review of portfolios of evidence and third-party workplace reports of on-the-job performance - Standardized assessment of character strengths and virtues applied	1 hour
	5.2 Discuss and develop ideas with others	- Identify different roles of individuals in contributing to doing things better in the workplace - Explain the concepts of positive impacts and challenges in innovation	- Interactive Lecture - Appreciative Inquiry - Demonstration	- Psychological and behavioral Interviews - Performance Evaluation	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		• Show mastery of the different types of changes and levels of participation in the workplace • Discuss 7 habits of highly effective people • Communicate ideas through small group discussions and meetings	• Group work	• Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	
	5.3 Integrate ideas for change in the workplace	• Identify different roles of individuals in contributing to doing things better in the workplace • Explain the concepts of positive impacts and challenges in innovation • Show mastery of the different types of changes and levels of participation in the workplace • Discuss 7 habits of highly effective people • Communicate ideas through small group discussions and meetings • Demonstrate basic skills in data analysis	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
6. Present relevant information	6.1 Gather data/ information	- Lecture and discussion on: - Organisational protocols - Confidentiality and accuracy - Business mathematics and statistics - Legislation, policy and procedures relating to the conduct of evaluations - Reviewing data/ information	- Group discussion - Lecture - Demonstration - Role Play	- Oral evaluation - Written Test - Observation - Presentation	2 hours
	6.2 Assess gathered data/ information	- Lecture and discussion on: - Data analysis techniques/ procedures - Organisational values, ethics and codes of conduct - Trends and anomalies - Computing business mathematics and statistics - Application of data analysis techniques	- Group discussion - Lecture - Demonstration - Role Play - Practical exercises	- Oral evaluation - Written Test - Observation - Presentation	3 hours
	6.3 Record and present information	- Lecture and discussion on: - Reporting requirements to a range of audiences - Recommendations for possible improvements - Analysis and comparison of interim and final reports' outcomes - Reporting of data findings	- Group discussion - Lecture - Demonstration - Role Play - Practical exercises	- Oral evaluation - Written Test - Observation - Presentation	3 hours
7. Practice Occupational Safety and Health Policies and Procedures	7.1 Identify OSH compliance requirements	- Discussion regarding: - Hierarchy of Controls - Hazard Prevention and Controls - Work Standards and Procedures - Personal Protective Equipment	- Lecture - Group Discussion	- Written Exam - Demonstration - Observation - Interviews / Questioning	1 hour
	7.2 Prepare OSH requirements for compliance	- Identification of required safety materials, tools and equipment - Handling of safety control resources	- Lecture - Group Discussion	- Written Exam - Demonstration - Observation - Interviews / Questioning	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	7.3 Perform tasks in accordance with relevant OSH policies and procedures	• Discussion of General OSH Standards and Principles • Performing industry related work activities in accordance with OSH Standards	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / Questioning	2 hours
8. Exercise Efficient and Effective Sustainable Practices in the Workplace	8.1 Identify the efficiency and effectiveness of resource utilization	• Identify the processes on environmental policies • Relate the necessary skills in response to changing environmental policies needs - Waste Management Skills - Conservation of Energy in workplace	• Lecture • Group Discussion • Simulation • Demonstration	• Written Exam • Demonstration • Observation • Interviews / Questioning	1 hour
	8.2 Determine causes of inefficiency and/or ineffectiveness of resource utilization	• Discussion of environmental protection and resource efficiency targets • Analysis on the relevant work procedure	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation • Interviews / Questioning	1 hour
	8.3 Convey inefficient and ineffective environmental practices	• Identification of (re)training needs and usage of environment friendly methods and technologies • Identification of environmental corrective actions • Practicing environment awareness	• Lecture • Group Discussion • Role Play • Demonstration	• Written Exam • Demonstration • Observation • Interviews / Questioning	1 hour
9. Practice Entrepreneurial Skills in the Workplace	9.1 Apply entrepreneurial workplace best practices	• Case studies on best entrepreneurial practices • Discussion on quality procedures and practices • Case studies on cost consciousness in resource utilization	• Case Study • Lecture/ Discussion	• Case Study • Written Test • Interview	1 hour
	9.2 Communicate entrepreneurial workplace best practices	• Discussion on communicating entrepreneurial workplace best practices	• Lecture/ Discussion	• Written Test • Interview	1 hour
	9.3 Implement cost-effective operations	• Case studies on Preservation, optimization and judicious use of workplace resources	• Case Study • Lecture/ Discussion	• Case Study • Written Test • Interview	2 hours

COMMON COMPETENCIES (45 HOURS)

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Update industry knowledge (11 hrs)	1.1 Seek and apply information on the industry	- Lecture and discussion on the following topics: - Industry information sources - Overview of quality assurance in the industry - Role of individual staff members - Perform the following tasks: - Identify and access sources of information on the industry - Obtain information to assist effective work performance in line with job requirements - Access and update specific information on sector of work - Correctly apply Industry information to day-to-day work activities - Demonstrate ready skills needed to access industry information - Demonstrate basic competency skills needed to access the internet	- Lecture-discussion - Demonstration - Hands-on/ Writeshop - Seminars/ conferences	- Written test - Oral Questioning - Presentation - Evaluation of written output	8 hours
	1.2 Update industry knowledge	- Lecture and discussion on the following topics: - Role of individuals in a creative endeavor member - Sources of Industry information - Perform the following tasks: - Use informal and/or formal research to update general knowledge of the industry - Share updated knowledge with customers and colleagues as appropriate and incorporate into day-to- day working activities - Demonstrate time management skills - Demonstrate ready skills needed to access industry information	- Lecture-discussion - Demonstration - Hands-on/ Writeshop - Research	- Written test - Oral Questioning - Presentation - Evaluation of written output	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
2. Manage own performance (8 hrs)	2.1. Plan for completion of own workload	- Lecture and discussion on the following topics: - Teamwork - Resource management - Timelines - Perform the following tasks: - Identify tasks are identified according to work instructions - Design and organize work order and schedules based on timelines/deadlines - Apply team coordination when required in completion of workload - Demonstrate planning and organizing workload and resources - Demonstrate communication skills	- Lecture - Group discussion - Demonstration - Hands-on/ Writeshop	- Written test - Oral Questioning - Presentation - Evaluation of written output	3 hours
	2.2. Maintain quality of performance	- Lecture and discussion on the following topics: - Indicators of appropriate performance for each area of responsibility - Steps for improving or maintaining performance - Perform the following tasks: - Monitor personal performance according to established plant standards - Obtain advice and guidance when necessary to achieve or maintain established standards - Apply guidance from management when necessary to achieve according to established standards	- Lecture - Group discussion - Demonstration - Hands-on/ Writeshop	- Written test - Oral Questioning - Presentation - Evaluation of written output	2 hours
	2.3. Evaluate and assess own work	- Lecture and discussion on the following topics: - Project Management - Process documentation - Perform the following tasks: - Evaluate and assess actual work output according to work instructions - Evaluate work productivity according to	- Lecture - Group discussion - Demonstration - Hands-on/ Writeshop	- Written test - Oral Questioning - Presentation - Evaluation of written output	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		established procedures ○ Obtain feedback from relevant persons ○ Manage work orders ○ Evaluate own performance ○ Report daily production activities			
3. Maintain safe, clean, and efficient work environment (13 hrs)	3.1. Comply with safety and health regulations	• Lecture and discussion on the following topics: ○ OSHS policies and standards ○ Fire code • Perform the following tasks: ○ Identify and comply with safety and health regulations ○ Adapt and apply policies and procedures ○ Comply with health and safety regulations ○ Demonstrate reading and comprehension	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	6 hours
	3.2. Assess work area	• Lecture and discussion on the following topics: ○ Work Hazards Policies and Procedures ○ OSHS policies and procedures ○ Waste management • Perform the following tasks: ○ Maintain work areas and walkways in a safe and hazard free environment ○ Carry out all routines in accordance with Occupational Safety and Health Standards (OSHS) ○ Store and dispose according to OSHS ○ Complying with health and safety regulations	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	5 hours
	3.3. Check and maintain tools, equipment and resources	• Lecture and discussion on the following topics: ○ Maintenance of tools and equipment ○ Tools, equipment and resource maintenance requirements • Perform the following tasks: ○ Store tools, equipment and resources	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		according to safety regulations ○ Check tools, equipment and resources for maintenance requirements ○ Refer tools and equipment for repair as required ○ Check for maintenance requirements ○ Store tools and equipment			
4. Perform basic computer operations (13 hrs)	4.1 Identify the basic components of a desktop publishing system	• Lecture and discussion on the following topics: ○ Basic components of a computer system ○ Computer specifications for desktop color publishing system ○ Various operating systems and platforms • Perform the following tasks: ○ Identify hardware of a computer according to desktop publishing system requirements ○ Identify software according to desktop publishing system requirements ○ Identify other peripherals according to desktop publishing system requirements ○ Identify kinds of computer components ○ Identify computer software ○ Identify computer peripherals	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	5 hours
	4.2 Select the appropriate hardware, software and other peripherals needed for desktop publishing	• Lecture and discussion on the following topics: ○ Appropriate hardware for desktop publishing ○ Appropriate software for desktop publishing • Perform the following tasks: ○ Select appropriate hardware and its specifications needed for desktop publishing ○ Select appropriate software needed for desktop publishing ○ Select peripherals needed for desktop publishing ○ Select appropriate hardware and software	• Lecture • Group discussion • Demonstration • Hands-on/ Writeshop	• Written test • Oral Questioning • Presentation • Evaluation of written output	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		Select appropriate peripherals for desktop publishing			
	4.3 Operate the basic components of a desktop publishing system	- Lecture and discussion on the following topics: - Desktop taskbars and icons - Local drive, partition drive and removable drive functions - Address bar and internet browsing - Networked computer connection - Perform the following tasks: - Access files in the server via network - Use appropriate software according to work instructions - Use properly other peripherals according to manufacturer's recommendation - Access files - Operate appropriate software - Demonstrate written and oral communications - Follow manufacturer's recommendation	- Lecture - Group discussion - Demonstration - Hands-on/ Writeshop	- Written test - Oral Questioning - Presentation - Evaluation of written output	3 hours
	4.4 Maintain the computer system and its external parts	- Lecture and discussion on the following topics: - Various types of virus and its effects on the computer - Various types of malicious software and its effects on the computer - File management system - Proper computer cleaning - Perform the following tasks: - Check computer regularly for any virus and malicious software - Clean external parts of the computer regularly - De-clutter computer desktop in accordance with file management procedures - Check of virus and malicious software - Remove virus that affects the system	- Lecture - Group discussion - Demonstration - Hands-on/ Writeshop	- Written test - Oral Questioning - Presentation - Evaluation of written output	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Recover files affected by malicious software ○ Clean the external parts of the computer			

CORE COMPETENCIES (200 HOURS)

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Perform halftone reproduction and its requirements (16 hours)	1.1 Identify halftone reproduction and its requirements	- Lecture and discussion on the following topics: - Image capturing devices preparation - Selection of appropriate software - Computer operation - Identification of halftone reproduction - Verbal and written communication - Perform the following tasks: - Identify file copy for halftone reproduction - Identify and prepare image capturing devices according to the task to be undertaken - Select appropriate software in accordance to specified reproduction requirements - Demonstrate effective communication skills - Determine differences between hard and soft copy - Operate different image capturing devices - Apply scanner software - Apply 6S principles	- Lecture - Group discussion - Demonstration - Case work	- Oral questioning/ interview - Written Test - Candidate's demonstration - Direct observation of candidate - Hands on	5 hours
	1.2 Convert a continuous tone copy into a printable image	- Lecture and discussion on the following topics: - Job specification - Client requirements - Flow of communication - Task planning and preparation - Information on characteristic of materials - Perform the following tasks: - Check materials to be reproduced if complete, fully identified & had complied according to work instructions	- Case work - Project method - Lecture - Group discussion	- Oral questioning/ interview - Written test - Return demonstration - Direct observation of candidate - Hands on	5 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Conducts checking on original copy to be reproduced ○ Set image capturing device settings according to established procedures ○ Capture hard copy in accordance to specified reproduction requirements ○ Gather client's specifications ○ Check client's requirements ○ Handle/keep supplied materials with care ○ Plan and prepare task			
	1.3 Recommend appropriate halftone dot frequency & dot shapes	• Lecture and discussion on the following topics: ○ Computer functions ○ Various software and their usage ○ Installation of identified various types of fonts • Perform the following tasks: ○ Determine paper-ink-press characteristics according to work instructions ○ Recommend appropriate halftone specifications in accordance with established procedures ○ Save a copy of reproduced image in an appropriate format for the next printing process ○ Demonstrate computer skills ○ Use different software related to printing ○ Identify and install various fonts	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Return demonstration • Direct observation of candidate • Hands on	6 hours
2. Perform basic color separation process (12 hours)	2.1 Check requirements for basic color separation process	• Lecture and discussion on the following topics: ○ Color Theory ○ Calibration of paper-ink-press system ○ Types of basic color separation ○ Types of images ○ Types of screen tint	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		- Perform the following tasks: - Determine basic color separation requirements according to work instructions - Determine drawing source according to work instructions - Review requirements for basic color separation of a drawing according to work instructions - Review work instructions - Demonstrate drawing skills		Direct observation of candidate	
	2.2 Prepare hardware and software necessary for basic color separation	- Lecture and discussion on the following topics: - Various color separation software - Various image capturing device operation - Color/spot color preparation - Perform the following tasks: - Perform digitization of artwork materials using image capturing device in accordance with established procedures - Set software in preparation for basic color separation process - Prepare necessary colors needed for basic color separation in accordance with work instructions - Digitize artwork materials - Apply color separation software - Demonstrate computer skills	- Case work - Project method - Lecture - Group discussion	- Oral questioning/ interview - Written test - Return demonstration - Direct observation of candidate	3 hours
	2.3 Perform basic color separation of an image	- Lecture and discussion on the following topics: - Layer creation - Archiving procedures - Selection and coloring tools - Concept of color separation - Perform the following tasks:	- Case work - Project method - Lecture - Group discussion	- Oral questioning/ interview - Written test - Return demonstration	4 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Create layers according to established procedures ○ Select areas to be applied with colors using the selection tools ○ Apply color screen tint, gradation and pattern into the selected areas using coloring tools ○ Save a copy for archiving purposes in accordance with established procedures ○ Layer digital file ○ Apply selection and coloring tools ○ Archive a copy ○ Demonstrate drawing/tracing techniques		• Direct observation of candidate	
	2.4 Print a copy for checking	• Lecture and discussion on the following topics: ○ Check of printed copy against work instruction ○ Application of correction and revision • Perform the following tasks: ○ Print a colored proof for checking in accordance with established procedures ○ Check the printed copy based on the original copy and work instructions ○ Forward the printed copy to relevant person for approval ○ Apply correction and revision to the proof submitted by relevant person ○ Print a proof ○ Check printed copy ○ Correct and revise file ○ Demonstrate communication skills	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate	3 hours
3. Perform digital output processing operations	3.1 Perform digital contract proofing operations	• Lecture and discussion on the following topics: ○ Functions of various proofing machines ○ Various types of proof ○ File handling	• Case work • Project method • Lecture	• Oral questioning/ interview • Written test	16 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
(80 hours)		○ Preflight procedures ○ Communication ○ Quality control procedures • Perform the following tasks: ○ Check digital printer if ready and in accordance with established procedures ○ Prepare printing requirements of a job according to work instructions ○ Print data file according to established procedures ○ Check the quality of printed proof according to established procedures ○ Forward checked proof to relevant personnel in accordance with established procedures ○ Understand work instructions ○ Prepare printing requirements ○ Check a file ○ Check the printing quality ○ Demonstrate oral and written communications	• Group discussion	• Return demonstration • Direct observation of candidate • Hands on	
	3.2 Perform plate recording and processing methods	• Lecture and discussion on the following topics: ○ Parts and functions of Computer-to-Plate equipment ○ Make-ready of plate making ○ Raster Image Processor functions ○ Plate quality inspections ○ Proper handling of consumables • Perform the following tasks: ○ Select computer-to-Plate (CtP) system according to work instructions ○ Prepare computer-to-Plate system according to established procedure ○ Review ready-file before sending to Raster Image Processor (RIP)	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	32 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Perform plate making make-ready according to established procedures ○ Examine ripped file against the contract proof according to established procedures ○ Perform imaging procedures according to established procedures ○ Inspect and forward plate to relevant persons for the next printing process ○ Prepare computer to plate system ○ Rip of data files ○ Review proof ○ Demonstrate 6's principles			
	3.3 Perform die-cutting guide for die-knife making output operations	● Lecture and discussion on the following topics: ○ Adobe Illustrator application ○ Mensuration ○ Mock-process procedures ● Perform the following tasks: ○ Determine source of die-cutting guide dimensions based on work instructions ○ Perform die-cutting guide making using Adobe Illustrator according to established procedures ○ Set line assignments according to work instructions ○ Save finished die-cutting guide layout in the appropriate format ○ Print finished die-cutting guide layout for inspections ○ Perform mocked-up process according to work instructions ○ Present the mock-up sample to relevant person for approval ○ Demonstrate measuring skills ○ Demonstrate drawing skills	● Case work ● Project method ● Lecture ● Group discussion	● Oral questioning/ interview ● Written test ● Return demonstration ● Direct observation of candidate ● Hands on	32 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Handle data ○ Printing of layout ○ Forming skills ○ Communication skills			
4. Perform imposition procedures (80 hours)	4.1 Create a master layout for publication	• Lecture and discussion on the following topics: ○ Computer application ○ Basic measurement ○ Verbal and written communication ○ Various types of folding dummy ○ Press and post-press requirements ○ Concept of imposition • Perform the following tasks: ○ Check pagination of folding dummy against soft file pagination ○ Create master layout using the appropriate software in accordance with work instructions ○ Set imposition measurements in accordance with established procedures ○ Demonstrate communication Skills ○ Demonstrate interpersonal skills ○ Operate computer ○ Perform imposition application	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate	24 hours
	4.2 Perform nesting of jobs	• Lecture and discussion on the following topics: ○ Procedures on producing packaging products ○ Concept of job nesting ○ Die-cutting knife function ○ Die-line interpretation ○ Quality control methods • Perform the following tasks: ○ Check design layout and die-cutting layout if all elements are complete according to work instructions	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	16 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Check die-cutting layout if it conforms with the size of the design layout ○ Compare layout with die-cutting guide against the mock-up sample provided by the client ○ Nest layout with or without die-cutting guide in accordance with work instructions ○ Check nested jobs to avoid printing errors according to established quality control procedures ○ Perform quality inspection ○ Nesting of jobs ○ Interpret work instructions ○ Apply line reproduction requirements			
	4.3 Insert required marks for press and post-press	• Lecture and discussion on the following topics: ○ Functions of marks ○ Marks for press requirements ○ Marks of post-press requirements ○ Information marks needed for archiving • Perform the following tasks: ○ Insert required marks for press according to press established procedures ○ Insert required marks for post-press according to post-press established procedures ○ Insert other information marks according to established procedures ○ Insert various types of marks ○ Position of marks	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	16 hours
	4.4 Perform manual / automatic imposition	• Lecture and discussion on the following topics: ○ Concept of manual imposition ○ Concept of electronic imposition ○ Archiving procedures • Perform the following tasks:	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration	20 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Imposed individual pages either manually or electronically ○ Review imposition according to established company standards ○ Save final imposed job in preparation for the next process ○ Impose manually ○ Impose electronically ○ Review imposed jobs		• Direct observation of candidate • Hands on	
	4.5 Print a small scale sample for checking	• Lecture and discussion on the following topics: ○ Advance printer's preference ○ Requirements for binding ○ Requirements for nesting • Perform the following tasks: ○ Print a small scale sample (25%) of the imposed job for final imposition check ○ Check folding sequence in accordance with the binding requirements ○ Arrange in order the folded pages in accordance with work instruction ○ Collate signatures ○ Insert signatures ○ Check printed copies	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	4 hours
5. Perform regular maintenance of condition-based prepress equipment / machines (12 hours)	5.1 Perform inspection on major parts of equipment/machines based on the agreement with the supplier	• Lecture and discussion on the following topics: ○ Parts of equipment/ machines ○ Equipment/machines consumables ○ Ink/water/air flow systems ○ Output quality standards ○ Various digital prepress machines ○ Supplier-client maintenance agreement • Perform the following tasks:	• Case work • Project method • Lecture • Group discussion	• Oral questioning/ interview • Written test • Return demonstration • Direct observation of candidate • Hands on	6 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		○ Perform inspection regarding the wear and tear of major parts of equipment/machines ○ Perform inspections on ink/water/air flow systems of equipment/machines for possible clogging and damage ○ Perform inspections on the output quality of various digital outputting machines ○ Inspect parts of prepress machines/equipment ○ Inspect output quality ○ Inspect ink/water/air flow systems			
	5.2 Perform fault finding check on the inspected parts of equipment/machines	• Lecture and discussion on the following topics: ○ Check on various prepress machines/equipment • Perform the following tasks: ○ Perform fault finding check on machine parts in accordance with established procedures ○ Record fault finding check in accordance with established procedures ○ Submit fault finding check reports to relevant person for further actions ○ Demonstrate communication skills ○ Demonstrate analytical skills	• Demonstration • Lecture • Group discussion	• Oral questioning/interview • Written test • Return demonstration • Direct observation of candidate • Hands on	6 hours

3.2 TRAINING DELIVERY

1. The delivery of training shall adhere to the design of the curriculum. Delivery shall be guided by the principles of competency-based TVET.
 - a. Course design is based on competency standards set by the industry or recognized industry sector; (Learning system is driven by competencies written to industry standards)
 - b. Training delivery is learner-centered and should accommodate individualized and self-paced learning strategies;
 - c. Training can be done on an actual workplace setting, simulation of a workplace and/or through adoption of modern technology.
 - d. Assessment is based in the collection of evidence of the performance of work to the industry required standards;
 - e. Assessment of competency takes the trainee's knowledge and attitude into account but requires evidence of actual performance of the competency as the primary source of evidence.
 - f. Training program allows for recognition of prior learning (RPL) or current competencies;
 - g. Training completion is based on satisfactory completion of all specified competencies not on the specified nominal duration of learning.
2. The competency-based TVET system recognizes various types of delivery modes, both on-and off-the-job as long as the learning is driven by the competency standards specified by the industry. The following training modalities and their variations/components may be adopted singly or in combination with other modalities when designing and delivering training programs:

2.1 Institution- Based:

- Dual Training System (DTS)/Dualized Training Program (DTP) which contain both in-school and in-industry training or fieldwork components. Details can be referred to the Implementing Rules and Regulations of the DTS Law and the TESDA Guidelines on the DTP.
- Distance learning is a formal education process in which majority of the instruction occurs when the students and instructor are not in the same place. Distance learning may employ correspondence study, audio, video, computer technologies; or other modern technology that can be used to facilitate learning and formal and non-formal training. Specific guidelines on this mode shall be issued by the TESDA Secretariat.
- Supervised Industry Learning (SIL) or on-the-job training (OJT) is an approach in training designed to enhance the knowledge and skills of the trainee through actual experience in the workplace to acquire

specific competencies as prescribed in the training regulations. It is imperative that the deployment of trainees in the workplace is adhered to training programs agreed by the institution and enterprise and status and progress of trainees are closely monitored by the training institutions to prevent opportunity for work exploitation.

- The classroom-based or in-center instruction uses of learner-centered methods as well as laboratory or field-work components.

2.2 Enterprise-Based:

- Formal Apprenticeship – Training within employment involving a contract between an apprentice and an enterprise on an approved apprenticeable occupation.
- Informal Apprenticeship - is based on a training (and working) agreement between an apprentice and a master craftsperson wherein the agreement may be written or oral and the master craftsperson commits to training the apprentice in all the skills relevant to his or her trade over a significant period of time, usually between one and four years, while the apprentice commits to contributing productively to the work of the business. Training is integrated into the production process and apprentices learn by working alongside the experienced craftsperson.
- Enterprise-based Training - where training is implemented within the company in accordance with the requirements of the specific company. Specific guidelines on this mode shall be issued by the TESDA Secretariat.

2.3 Community-Based

- Community-Based – short term programs conducted by non-government organizations (NGOs), LGUs, training centers and other TVET providers which are intended to address the specific needs of a community. Such programs can be conducted in informal settings such as barangay hall, basketball courts, etc. These programs can also be mobile training program (MTP).

3.3 TRAINEE ENTRY REQUIREMENTS

Trainees or students wishing to enroll in this program must possess the following requirements:

- Completed at least grade 10 of basic education or holder of Alternative Learning Systems (ALS) certificate of completion with grade 10 equivalent; and
- Must possess good communication skills

3.4 LIST OF TOOLS, MATERIALS AND EQUIPMENT

Recommended list of tools, equipment and materials for the training of **25 trainees** for Printing Services (Prepress Technical Operations) NC II.

Up-to-date tools, materials, and equipment of equivalent functions can be used as alternatives. This also applies in consideration of community practices and their availability in the local market.

TOOLS

QTY	UNIT	DESCRIPTION/SPECIFICATION
26	pcs	USB Flash Drives, 8G
26	pcs	Magnifying Lens (10x)
26	pcs	Steel Rule, 24"
1	pcs	Whiteboard (8' x 4')
1	unit	Room Temperature and Humidity Meter (Digital)

EQUIPMENT

QTY	UNIT	DESCRIPTION/SPECIFICATION
26	units	Work station computer unit with accessories and installed desktop publishing software Computer Specifications: • Processor: at least i5 or higher (at least 10th generation or higher) • RAM: minimum of 8gb 16gb • Video Card: minimum of 4gb • Storage: minimum of 256gb (SSD) • Monitor: minimum of 24" LED • Compatible Platform • UPS – 650 VA Software Specifications: • Lay –out Designing Software • Vector-Based Software • Image Manipulation and Editing Software • Productivity Software Tools or Word Processing Software
5	units	Flatbed Scanner with OCR Software DPI 1200, A3 size
21	units	LCD Projector
21	units	72" LED Monitor Calibrated
1	unit	Monitor Calibrator
1	unit	Computer set as file server • Processor: at least i5 or higher (at least 10th generation or higher) • RAM: minimum of 8gb

		- Storage: 1tb as Operating System 1tb 2tb as Main Storage 1tb as Mirrored Disk - UPS – 650 VA Monitor: 24”
1	unit	Audio System with Lapel Microphone
1	unit	QR Code Scanner
1	unit	Laser Desktop Printer, Standard
1	unit	Inkjet Desktop Printer, Standard
1	unit	Plate Recording Unit, 4 ups 32 diodes with UPS
1	unit	Plate Processor, 4 ups
1	unit	Plate Viewing Table, 4 ups

MATERIALS

QTY	UNIT	DESCRIPTION/SPECIFICATION
26	pcs	Ballpen
26	pcs	Pencil
26	pcs	Hard Case Folder with 3 rings
26	pcs	Fastener
26	pcs	Notebook
1	pc	Log Book
26	pcs	IDs with QR code
5	pcs	1” Camel’s Hairbrush
3	reams	Copy Paper (Letter, A4 & Legal)
3	pcs	Whiteboard Marker (Black, Blue and Red)
1	pc	Whiteboard Eraser
26	pcs	Face Shield
26	pcs	Surgical Mask (disposable)
26	pcs	Sponge
26	pcs	Latex Gloves
26	pcs	Plastic Apron
26	pcs	Round Rugs
1	bar	Hand Soap
1	bottle	70% Isopropyl Alcohol (500ml)
1	bottle	Dishwashing Liquid (500 ml)
1	bottle	Wipe n Shine solution
1	unit	Blower
1	unit	Light Duty Vacuum Cleaner

INSTRUCTIONAL MANUALS/FORMS

QTY	UNIT	DESCRIPTION/SPECIFICATION
		Standard Forms:
26	copies	Performance Evaluation Form
26	copies	Variable Data Information Sheet
26	copies	Inspection Schedule Form
26	copies	Inspection Checklist Form
26	copies	Routine Maintenance Report
26	copies	Parts Replacement Report
5	copies	Sample Manuscript
5	copies	Sample Printed Copy for OCR Scanning
5	copies	Printed or Reflective Copies
5	copies	Artwork Materials
		Relevant Policies and Procedure Manuals:
26	copies	Preflighting Procedures Manual
26	copies	File Handling Procedure Manual
26	copies	Folder Structure Sample Manual
26	copies	Input Processing Procedure Manual
26	copies	Scanner's Operating Procedures
26	copies	Inkjet and Laser Printer's Operating Procedures
26	copies	Line Reproduction Procedures Manual
26	copies	Digital Artwork Tracing Procedures
26	copies	Plate Recorder's Operating Procedure
26	copies	Plate Processor's Operating Procedure
		Instructional Materials
26	pcs	Contract Proof Sample
26	sheets	Preflighting Checklist
26	sheets	Preflighting Report Template
26	sets	Various Printed Samples
26	copies	Proofreading Marks/Symbols Handouts
26	copies	Work Instruction Form
26	copies	Job Order form
26	copies	Task Sheet
26	copies	Job Sheet
26	copies	Theoretical Lecture Handouts
5	pcs	Die-cutting Form Sample

3.5 TRAINING FACILITIES

Based on a class intake of 25 learners/trainees.

Space Requirement	Size in Meters	Area in Sq. Meters	Total Area in Sq. Meters
A. Distance Learning (Laboratory/Workshop/Activity area)	12 x 15	180	180
B. Practical Work Area	6 x 6	36	36
C. Contextual Learning Area (Lecture Room)	10 x 12	120	120
D. Store/Storage Room	2 x 5	10	10
E. Separate restrooms for female, male and PWD		10	10
F. Circulation Area		22	22
Total Workshop Area:			378 sq. m.

3.6 TRAINER'S QUALIFICATIONS

- Must be a holder of National TVET Trainer Certificate (NTTC) Level I in Printing Services (Prepress Technical Operations) NC II OR must be a practicing trainer for two (2) years within the last five (5) years;
- Must have at least three (3) years prepress technical operations-related industry experience¹ within the last five (5) years **OR** must be a practicing trainer/with teaching experience² related to prepress technical operations for two (2) years within the last five (5) years; and
- Must possess good communication skills

¹ A certificate of employment or any document which will serve as evidence that indeed the trainer has industry experience must be submitted.

² Refer to the existing provision of INDUSTRY WORK EXPERIENCE REQUIRED (IWER)

3.7 INSTITUTIONAL ASSESSMENT

Institutional assessment is undertaken by trainees to determine their achievement of units of competency. A certificate of achievement is issued for each unit of competency.

SECTION 4 ASSESSMENT AND CERTIFICATION ARRANGEMENT

Competency Assessment is the process of collecting evidence and making judgments whether competency has been achieved. The purpose of assessment is to confirm that an individual can perform to the standards expected at the workplace as expressed in relevant competency standards.

The assessment process is based on evidence or information gathered to prove achievement of competencies. The process may be applied to an employable unit(s) of competency in partial fulfillment of the requirements of the national qualification.

4.1 NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

4.1.1 To attain the national qualification of **PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II**, the candidate must demonstrate competence in all units listed in Section 1. Successful candidates shall be awarded a National Certificate signed by the TESDA Director General.

4.1.2 Assessment shall cover all competencies with basic and common integrated or assessed concurrently with the core units of competency.

4.1.3 Any of the following are qualified to undergo assessment and certification:

4.1.3.1 Graduates of WTR-registered, NTR-registered programs or formal/non-formal/informal including enterprise-based trainings related to Printing Services (Prepress Technical Operations) NC II; or

4.1.3.2 Experienced workers (wage employed or self-employed) who gained competencies in providing prepress technical operations services for at least two (2) years within the last five (5) years.

4.1.4 **Recognition of Prior Learning (RPL).** Candidates who have gained competencies through education, informal training, previous work or life experiences in prepress technical operations for at least three (3) years within the last five (5) years may apply for recognition in this Qualification through Portfolio Assessment.

Requirements and implementation procedure of Portfolio Assessment must be consistent with TESDA Circular No. 025, Series of 2023 entitled “*Amended Operating Procedure on the Conduct of Portfolio Assessment*”.

4.1.5 The guidelines on assessment and certification are discussed in detail in the “Procedures Manual on Assessment and Certification” and “Guidelines on the Implementation of the “Philippine TVET Competency Assessment and Certification System (PTCACS)”.

4.2 COMPETENCY ASSESSMENT REQUISITE

4.2.1 **Self-Assessment Guide.** The self-assessment guide (SAG) is accomplished by the candidate prior to actual competency assessment. SAG is a pre-assessment tool to help the candidate and the assessor determine what evidence is available, where gaps exist, including readiness for assessment.

This document can:

- a) Identify the candidate's skills and knowledge
- b) Highlight gaps in candidate's skills and knowledge
- c) Provide critical guidance to the assessor and candidate on the evidence that need to be presented
- d) Assist the candidate to identify key areas in which practice is needed or additional information or skills that should be gained prior to assessment.

4.2.2 **Accredited Assessment Center.** Only Assessment Center accredited by TESDA is authorized to conduct competency assessment. Assessment centers undergo a quality assured procedure for accreditation before they are authorized by TESDA to manage the assessment for National Certification.

COMPETENCY MAP – SOCIAL AND OTHER COMMUNITY DEVELOPMENT SERVICES SECTOR PRINTING SERVICES (PREPRESS TECHNICAL OPERATIONS) NC II

BASIC COMPETENCIES

Receive and respond to workplace communication	Work with others	Solve/address routine problems	Enhance self-management skills	Support Innovation	Access and maintain information	Follow occupational safety and health policies and procedures	Apply environmental work standards	Adopt entrepreneurial mindset in the workplace
Participate in workplace communication	Work in team environment	Solve/address general workplace problems	Develop career and life decisions	Contribute to workplace innovation	Present relevant information	Practice occupational safety and health policies and procedures	Exercise efficient and effective sustainable practices in the workplace	Practice entrepreneurial skills in the workplace
Lead workplace communication	Lead small teams	Apply critical thinking and problem-solving techniques in the workplace	Work in a diverse environment	Propose methods of applying learning and innovation in the organization	Use information systematically	Evaluate occupational safety and health work practices	Evaluate environmental work practices	Facilitate entrepreneurial skills for micro-small-medium enterprises (MSMEs)
Utilize specialized communication skill	Develop and lead teams	Perform higher order thinking processes and apply techniques in the workplace	Contribute to the practice of social justice in the workplace	Manage innovative work instructions	Manage and evaluate usage of information	Lead in improvement of Occupational Safety and Health Program, Policies and Procedures	Lead towards improvement of environmental work programs, policies and procedures	Sustain entrepreneurial skills
Manage and sustain effective communication strategies	Manage and sustain high performing teams	Evaluate higher order thinking skills and adjust problem solving techniques	Advocate strategic thinking for global citizenship	Incorporate innovation into work procedures	Develop systems in managing and maintaining information	Manage implementation of occupational safety and health programs in the workplace	Manage implementation of environmental programs in the workplace	Develop and sustain a high-performing enterprise

**COMMON
COMPETENCIES**

Maintain instruments and equipment in work area	Assist in dental laboratory procedures	Assist with administration in dental laboratory practice	Implement and monitor infection control policies and procedures	Respond effectively to difficult/ challenging behavior	Apply basic first aid	Maintain high standard of patient / client services	Apply quality standards	Maintain an effective relationship with customers and clients	Maintain an effective relationship with clients/ customers (marketing)
Update industry knowledge	Manage own performance	Maintain a safe, clean and efficient environment	Perform computer operations	Follow occupational health and safety policies in dental laboratory facilities	Maintain infection control in dental practice	Operate a personal computer	Perform workplace security and safety practices	Update industry knowledge and practice through continuing education	Use pharmaceutical calculation techniques and terminologies
Maintain instruments and equipment in work area	Assist in dental laboratory procedures	Assist with administration in dental laboratory practice	Implement and monitor infection control policies and procedures	Respond effectively to difficult/ challenging behavior	Apply basic first aid	Maintain high standard of patient / client services	Apply quality standards	Maintain a safe, clean and efficient environment	Maintain an effective relationship with clients/ customers (marketing)

**CORE
COMPETENCIES**

Perform facial cleansing	Perform temporary hair removal activity	Perform body scrub	Perform pre and post hair care activities	Perform hair and scalp treatment	Perform basic hair coloring	Perform basic hair bleaching	Perform basic hair perming	Perform hair straightening
Perform basic haircutting	Perform advanced and creative haircutting	Perform advanced and creative hair coloring	Perform advanced and creative hair perming	Perform basic men's haircutting	Perform shave and style beard and mustache	Perform chair manipulative relaxing services	Perform manicure and pedicure	Perform hand and foot spa
Perform preparatory activities	Prepare appropriate products, tools and equipment	Perform nail enhancement technology procedures	Perform post service activities	Perform advanced nail polish procedures	Apply facial make-up	Perform body bleach	Network with stakeholders	Process labor market information
Market and promote services of public employment services office	Perform job matching services	Perform referral and placement services	Perform monitoring of jobseekers	Perform livelihood/training referral and facilitation services	Perform monitoring of referred livelihood and training applicants	Assist in the conduct of career coaching and employment orientation seminar to school-based clients	Provide employment orientation and coaching to walk-in clients	Provide employment orientation and coaching to other clients

Participate in the conduct of research and development of new programs	Implement employment facilitation services and special programs	Organize recruitment activities	Clean living room, dining room, bedroom, bathroom and kitchen	Wash and iron clothes, linen and fabric	Prepare hot and cold meals	Provide food and beverage service	Perform preparatory activities for data collection	Gather data
Perform post-data collection activities	Perform pre-flight procedures	Perform file handling	Perform input processing procedures	Perform line reproduction and its requirements	Perform preventive maintenance of prepress equipment/machines	Perform halftone reproduction and its requirements	Perform basic color separation process	Perform digital output processing operations
Perform imposition procedures	Perform regular maintenance of condition-based prepress machines/equipment							

GLOSSARY OF TERMS

Color Bars	Rows of different colored patches printed in the trim area of the press sheet. They are used by proofers and press operators to control the trapping, ink density, dot gain, and print contrast of the proof or the printed sheet.
Color Theory	Is both the science and art of using color. It explains how humans perceive color; and the visual effects of how colors mix, match or contrast with each other.
Computer-to-Plate	An imaging technology which helps to transfer a digital image generated in a computer directly to a printing plate.
Contract Proof	Serves as an agreement between customer and printer and as a color reference guide for adjusting the press before the final press run.
Creasing	A ridge or groove produced in anything by folding, heat, pressure, etc.
Die-Cutting	The process of cutting a piece of paper or material to a custom shape and size.
Digitization	The process of converting information into a digital format.
Folding Dummy	Sheet or sheets which has been folded into pages and marked with page numbers to indicate the proper page imposition. It also indicates where the gripper margin is.
Gradation	A visual technique of gradually transitioning from one color hue to another, or from one shade to another, or one texture to another.
Gripper	A space on a sheet of paper that allows for the press to physically grip the paper. This is where the machine pulls the paper and guides it along the path to print.
Halftone Frequency	Refers to how many rows, or lines, of dots fit in a linear inch. The number of LPI in a halftone screen is called the screen frequency.
Imposition	The positioning of pages on a press sheet in such a manner that when the sheet is folded into a signature and cut, the pages will be in the correct sequence.
Make-Ready	The act or process of making something ready for use.
Nesting	The layout strategy that allows printers to organize the shapes of products to be cut out by nesting them in one another so as to optimize the use of space on the sheet.
Perfect Binding	A form of bookbinding in which the leaves are bound by gluing, after the back folds have been cut off, rather than by sewing.
Perforation	A small hole or row of small holes punched in a sheet of paper
Plate Control Wedge	Provides a means of optimizing the exposure of direct digital platemakers by visual inspection.

Printing Signature	A group of pages that are printed, most likely on both sides of a single sheet of paper that once folded, trimmed, bound and cut, become a specific number of pages.
Prototype	An early sample, model, or release of a product built to test a concept or process. A prototype is generally used to evaluate a new design to enhance precision by system analysts and users.
Raster Image Processing	A component used in a printing system which produces a raster image also known as a bitmap. Such a bitmap is used by a later stage of the printing system to produce the printed output.
Saddle Stitching	A very popular book binding method in which folded sheets are gathered together one inside the other and then stapled through the fold line with wire staples.
Smyth Sewn Binding	Method of book binding where groups of folded pages (referred to as signatures) are stitched together using binder thread. Each folded signature is sewn together individually with multiple stiches and then joined with other signatures to create the complete book block.
Step and Repeat	The process of creating a layout where an image, logo or sponsor name is repeated multiple times, creating a uniform pattern.
Substrate	Term for any surface to be printed to which ink will adhere. The substrate, also called stock, is typically a paper, but can also refer to plastics, foil, metal, cloth, or any other surface to which printing ink will be applied.
Tonal Correction	The adjustments and changes you make to the brightness and contrast of your image.
Web-Press	A modern offset printing press attached to a massive roll of paper.

ACRONYMS

AI	Adobe Illustrator file extension
CDR	CorelDRAW file extension
CMS	Color Management System
CMYK	Cyan, Magenta, Yellow, Black
CtP	Computer-to-Plate
DVI	Digital Visual Interface
EPS	Encapsulated PostScript file extension
GCR	Gray Component Replacement
HDMI	High-Definition Multimedia Interface
INDD	Adobe InDesign file extension
JO	Job Order
JPEG	Joint Photographic Experts Group
LED	Light-Emitting Diode
OCR	Optical Character Recognition
OHS	Occupational Health and Safety
PDF	Portable Document Format
PS	Postscript
PSD	Photoshop Document file extension
RAM	Random-Access Memory
RGB	Red, Green, and Blue
RIP	Raster Image Processor
TIFF/TIF	Tagged Image File Format file extension
UCR	Under Color Removal
USB	Universal Serial Bus
VGA	Video Graphics Array

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